Resolution-Difference Embedded Network for Cross-Resolution Remote Sensing Image Change Detection

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***Abstract*—At present, most remote sensing image change detection (CD) methods are applicable to equal-resolution scenarios, that is, the bitemporal images are assumed to have the same spatial resolution. Real-world tasks such as disaster emergency response have put forward the need for a CD of bitemporal images with different spatial resolutions, that is, a cross-resolution CD. However, due to the significant differences in spatial details between high-resolution (HR) images and low-resolution (LR) images, it is difficult to extract the spatial features of changed landcovers in complex scenarios and distinguish between changed landcovers and unchanged landcovers. To overcome the above issues, we propose a resolution-difference embedding network (RDENet). RDENet combines two innovative methods: pseudo-continuous resolution sequence representation (PRSR) and bitemporal resolution-difference modulation (BRDM). The PRSR method effectively extracts features of landcovers in complex scenarios by constructing a pseudo-image sequence with a smooth transition of resolution and developing a resolution-guided feature fusion (RGFF) module. The BRDM method significantly enhances the model's ability to distinguish between changed and unchanged landcovers in complex scenarios through the design of a resolution-aware self-modulation (RASM) module and resolution-aware mutual-modulation (RAMM) module, which utilizes the resolution-difference factor as prior knowledge to dynamically enhance key features of changed landcovers in cross-resolution image pairs. Extensive experiments conducted on three publicly available CD datasets demonstrate that the proposed RDENet achieves superior detection performance in cross-resolution scenarios.**

***Index Terms*—Change detection (CD), cross resolution, deep learning, feature interaction, pseudo-image sequence.**

I. Introduction

REMOTE sensing change detection (CD) is the task of automatically identifying changed landcovers by comparing remote sensing images of the same geographical area captured at different times [1], [2], [3]. CD can analyze the changes of landcovers and is widely used in various fields, e.g., urban planning and management, disaster monitoring and assessment, environmental protection, and ecological monitoring [4], [5], [6].

Currently, deep learning techniques have been widely applied to CD tasks due to their powerful feature extraction capabilities [7], [8]. However, in practical applications, such as disaster emergency response, remote sensing images acquired at different times often come from satellites with varying resolutions, i.e., cross-resolution CD tasks. For cross-resolution CD, due to the differing spatial resolutions between bitemporal remote sensing images, spatial details of changed landcovers in complex scenes are prone to be lost. Moreover, the mixed pixel effect in low-resolution (LR) image exacerbates the radiometric discrepancies between the bitemporal images, and the impacts of shadow interference and noise are also amplified. These factors make it challenging for conventional deep learning models to adaptively extract discriminative features across resolutions [9], [10]. Most existing CD methods assume that bitemporal remote sensing images possess the same spatial resolutions, lacking specific designs for handling images with different resolutions. Consequently, these methods demonstrate poor performance in cross-resolution CD tasks [9], [10].

The primary approaches to addressing cross-resolution CD fall into two categories: image-space alignment and feature-space alignment. Image-space alignment aligns bitemporal images through upsampling, applying super-resolution methods to LR images [9], [11], [12], [13]. In contrast, feature-space alignment focuses on learning resolution-insensitive embedding spaces through interactive learning between features of different resolutions [10].

However, existing cross-resolution CD methods have limitations in identifying changed landcovers in complex scenarios. As shown in Fig. 1, under the ninefold resolution difference, in the red box of the CD results, the advanced image-space alignment method, super-resolution-based CD network (SRCDNet) [9], and feature-space alignment method, scale-invariant learning with implicit neural network (SILI) [10], are unable to detect changed vehicles affected by shadow occlusion, resulting in missing detections. In order to analyze the causes of missing detections, the features of the model output are visualized. The feature visualization is based on the Grad-CAM method [14], which extracts features

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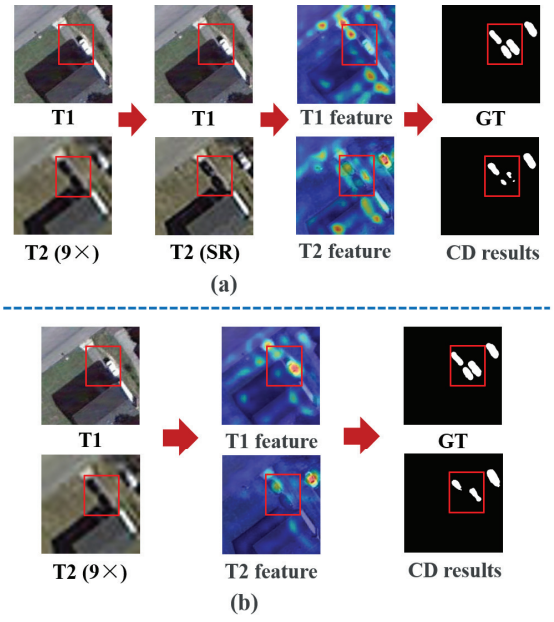


Fig. 1. (a) Image-space alignment method SRCDNet and (b) feature-space alignment method SILI achieved poor detection results for shadow-affected vehicle targets under complex remote sensing scenarios with a resolution difference of 9×. (SR denotes super-resolution, GT denotes ground truth, and CD denotes change detection.)

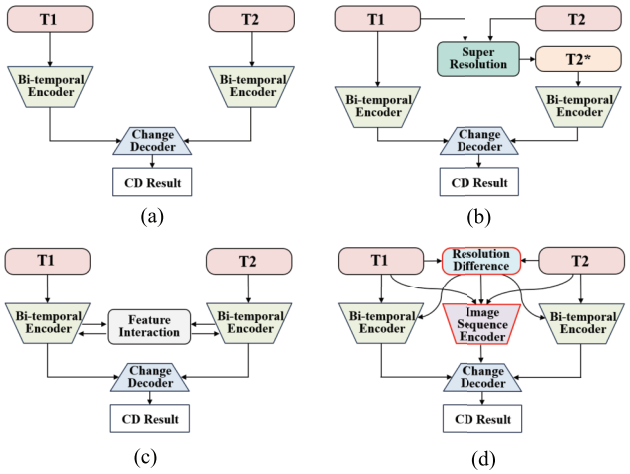


Fig. 2. Architecture diagrams of conventional CD method and several cross-resolution CD methods (taking the T1-temporal as HR image and the T2-temporal as LR image as an example). (a) Conventional CD method. (b) Image-space alignment method. (c) Feature-space alignment method. (d) Proposed RDENet.

from the last convolutional layer of the backbone network and generates activation values through $L2$-norm across the channel dimension, resulting in the final heatmap visualization. It can be observed that the vehicle being miss detected is not perceived in the feature map, making it difficult to distinguish between the changed and unchanged regions. Therefore, it is difficult for the existing cross-resolution CD method to effectively perceive the changed landcovers in bitemporal images, consequently failing to identify changed landcovers in complex scenarios.

In this article, a resolution-difference embedding network (RDENet) is proposed to enhance the model's perception capability of changed landcovers in cross-resolution remote sensing images, thereby achieving high-performance CD results. The network comprises two key components: a pseudo-continuous resolution sequence representation (PRSR) method and a bitemporal resolution-difference modulation (BRDM) method. The comparison between the proposed method and existing methods is illustrated in Fig. 2. To address the challenge of extracting spatial features of changed landcovers, PRSR is proposed. PRSR constructs a pseudo-resolution image sequence with smooth transitions, where spatial detail differences between adjacent images remain subtle. Through the pseudo-resolution-guided feature fusion mechanism, PRSR realizes the progressive fusion of adjacent cross-resolution features, which effectively mitigates the spatial feature extraction difficulties caused by significant resolution discrepancies in original images and significantly improves the extraction of discriminative spatial features of changed landcovers in complex scenarios. Meanwhile, in order to enhance the key features of the changed landcovers in the cross-resolution image for effective identification, the BRDM method is proposed. BRDM incorporates the resolution difference of bitemporal images into the learning paradigm through a resolution-difference factor embedded feature modulation mechanism. This mechanism utilizes the resolution-difference factor as prior knowledge to dynamically weight key features of changed landcovers in cross-resolution image pairs, thereby improving discriminability between changed and unchanged landcovers in complex remote sensing scenarios.

Different from the existing methods, PRSR constructs a pseudo-image sequence with continuous transitions, simulating the dynamic evolution of landcovers, and enables feature modeling in both temporal and spatial dimensions, thereby assisting the model in more accurately localizing changed landcovers. Moreover, the proposed BRDM innovatively incorporates the resolution difference between the bitemporal images as prior information into the feature extraction process. It adaptively generates key feature weights, thereby enhancing the key features of landcovers and facilitating their accurate identification. The main contributions of this article can be summarized as follows.

1) This article proposes an RDENet, incorporating PRSR and BRDM methods. These methods enhance the model's ability to perceive changed landcovers in cross-resolution images of complex scenarios, achieving high-performance CD under cross-resolution conditions.
2) To address the difficulty of extracting spatial features from cross-resolution images in complex scenarios, this article proposes a PRSR method. PRSR constructs a pseudo-image sequence with a smooth transition of resolution, and the adjacent images in the sequence have small spatial detail differences. Through the designed RGFF module, the features of adjacent pseudo-images are gradually fused to effectively capture the spatial features of landcovers and improve the model's ability to identify changed landcovers.
3) To enhance the key features of changed landcovers in cross-resolution images for distinguishing between

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changed and unchanged landcovers, a BRDM method is proposed. BRDM incorporates a resolution-difference factor embedded feature modulation mechanism, which utilizes the resolution-difference factor as prior knowledge to dynamically weight key features of changed landcovers in cross-resolution image pairs, thereby significantly improving the model's capability to differentiate changed and unchanged landcovers in complex remote sensing scenarios.

The remainder of this article is organized as follows. Section II provides a comprehensive review of existing methodologies. Section III elaborates the details of the proposed RDENet. Section IV presents extensive experimental evaluations. Section V systematically performs ablation studies. Finally, Section VII concludes this article.

II. Related Works

This section briefly introduces the works related to the proposed method from two aspects: CD and cross-resolution CD.

A. Change Detection

With the continuous development of deep learning technology and the emergence of labeled remote sensing datasets, deep learning-based CD methods have become the mainstream direction of CD research due to their powerful feature representation capabilities [15], [16], [17], [18]. Currently, most methods adopt an encoder–decoder deep learning framework, extracting features from bitemporal images and then fusing these features to obtain CD results. A representative work is the fully convolutional Siamese network proposed by Daudt et al. [19], and most subsequent deep learning-based CD methods are improvements based on this.

Recent deep learning-based CD research can be categorized into Transformer-based methods and CNN-based methods according to different backbone networks [20], [21]. Transformers can model long-range semantic dependencies of landcovers in remote sensing images [22], [23], [24], [25], [26], but may lack inductive biases [27]. Consequently, the vast majority of current methods still focus on CNN-based methods.

Regarding CNN-based methods, many methods have incorporated biologically inspired attention mechanisms to enhance features relevant to changed region identification, thereby achieving superior detection performance [28]. For instance, Zhang et al. [29] utilized an attention module to effectively fuse the deep features of the original images with the bitemporal difference features, thereby achieving accurate CD results. Chen et al. [30] proposed a dual attention mechanism for CD, which captures the long-range dependencies of landcovers to obtain discriminative feature representations, thereby improving the recognition performance of the model. Feng et al. [31] proposed a dual-branch multilevel intertemporal network (DMINet) that combines self-attention and cross attention to guide global feature distribution, stimulate information coupling between internal representations, and suppress task-independent interference. Shu et al. [32] proposed the prior guidance and principal attention network (PGPANet), which includes the designed principal attention module. This module aggregates global information from autonomous regions through sparse operations, filling the gap in global information in upsampling features and narrowing the semantic gap between the encoder and decoder. Pan et al. [33] proposed a spatial selection and temporal evolution network (STENet), which adopts a dynamic selective attention mechanism to ensure mild computational requirements while striving for spatial feature interaction and attention to changed regions.

It is worth noting that the aforementioned methods mainly address CD tasks for remote sensing images with identical spatial resolutions. In the real world, with the continuous launching of high-, medium-, and low-orbit satellites, remote sensing satellites are evolving toward a constellation-based development trend, leading to a significant increase in resolution difference among remote sensing images [34], [35]. For remote sensing applications, such as disaster emergency response, the utilization of multiresolution images enables rapid and effective acquisition of landcover change information, thereby imposing stringent demands on cross-resolution CD tasks [10]. However, the above-mentioned CD methods are challenged by resolution differences and have poor performance in solving cross-resolution CD tasks. Therefore, there is an urgent need to study cross-resolution CD methods.

B. Cross-Resolution CD

Cross-resolution CD, which can extract landcover change information by processing remote sensing images with different spatial resolutions, has attracted increasing attention from scholars in the remote sensing community due to its applicability across various fields [9], [10], [36], [37], [38]. This article primarily focuses on cross-resolution tasks for optical remote sensing images.

A straightforward solution is to upsample LR images to match the resolution of high-resolution (HR) images, thereby aligning bitemporal images in the image space [9], [11], [12]. For example, Liu et al. [9] proposed an SRCDNet. SRCDNet adopts a super-resolution method that includes a generator and a discriminator and directly learns LR to HR image mapping through adversarial learning, overcoming the resolution difference between bitemporal images. However, the image-space alignment method neglects the textural details of landcovers in LR images, and spatial detail discrepancies between bitemporal images persist, making it challenging to achieve accurate detection results.

In contrast to image-space alignment, recent studies have focused on exploring feature-space alignment methods [37], [39]. Feature-space alignment methods achieve cross-resolution semantic consistency by learning the mapping relationships between LR image features and HR image features. Chen et al. [10] proposed a SILI to solve cross-resolution CD tasks. This method considers that spatial resolution mainly affects local texture and applies local-window self-attention to align bitemporal features in the early stages of the encoder. However, current feature-space alignment methods demonstrate insufficient utilization of inherent bitemporal

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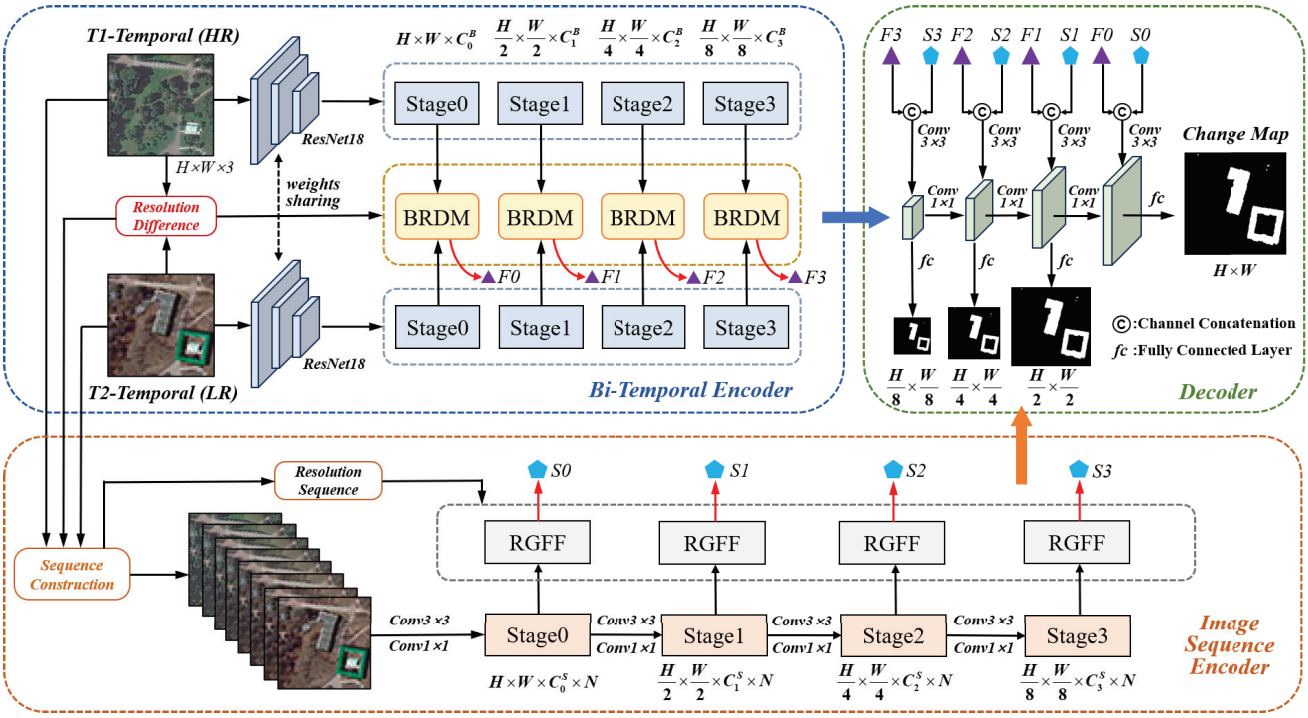


Fig. 3. Overall architecture of RDENet. RDENet comprises two core components: PRSR and BRDM. PRSR method constructs a pseudo-image sequence with continuously transitioning resolutions and extracts cross-resolution spatiotemporal features from it. The BRDM method dynamically captures key features across different resolutions, enabling adaptive enhancement of cross-resolution features. (A pretrained ResNet18 is used as the Siamese feature extractor, BRDM stands for bitemporal resolution-difference modulation, and RGFF stands for resolution-guided feature fusion.)

image information, leading to compromised cross-resolution interaction efficacy between bitemporal features.

Different from existing image-space and feature-space alignment methods, this article addresses cross-resolution CD challenge through dual perspectives of image and feature representation. From the image perspective, we construct multitemporal image sequences with smooth resolution transitions to mitigate the adverse impacts caused by significant spatial detail discrepancies in bitemporal images on CD accuracy. From the feature perspective, explicit incorporation of the resolution-difference factor into the feature extraction pipeline enables the model to acquire discriminative feature representation capabilities across varying resolutions, effectively bridging the resolution gap in feature space.

III. Resolution-Difference Embedding Network

In the proposed RDENet, two branches are employed to extract landcover features. One branch is an image sequence encoder, which first constructs a pseudo-image sequence with continuous resolution transitions and then extracts spatiotemporal landcover features using pseudo-resolution-guided multitemporal feature fusion. The other branch utilizes a pretrained ResNet18 as a Siamese network to extract bitemporal features, followed by an attention mechanism embedded with a resolution-difference factor to enhance the key features. The features from both branches are fused through a decoder, ultimately yielding the CD results in cross-resolution scenarios.

A. Overview of RDENet

The proposed RDENet extracts cross-resolution landcover features through an image sequence encoder and a bitemporal encoder. The image sequence encoder performs RGFF on the constructed pseudo-continuous resolution image sequence. The bitemporal encoder contains a resolution-aware self-modulation (RASM) module and a resolution-aware mutual-modulation (RAMM) module to enhance cross-resolution key features. The outputs of the two encoders are fed into a decoder to obtain CD results.

The overview architecture of the proposed RDENet is shown in Fig. 3. Assume that $I_{\mathrm{HR}} \in \mathbb{R}^{H\times W\times 3}$ and $I_{\mathrm{LR}} \in \mathbb{R}^{H_L\times W_L\times 3}$ are used to represent the input HR image and LR image, respectively. The resolution-difference factor can be expressed as follows:

$$r = H_L/H. \tag{1}$$

To facilitate subsequent model computation, the LR image I_{LR} is first upsampled to the size of the HR image I_{HR} using bicubic interpolation

$$I^u_{\mathrm{LR}} = \text{upsampling}\left(I_{\mathrm{LR}}, r\right). \tag{2}$$

The proposed RDENet incorporates two specialized encoders for cross-temporal feature extraction: an image sequence encoder and a bitemporal encoder. For the image sequence encoder, the initial process is to construct a pseudo-image sequence with continuously transitioning resolutions from bitemporal images through interpolation

$$\left\{I^s_1, \ldots, I^s_N\right\}, \{R_1, \ldots, R_N\} = \Psi\left(I_{\mathrm{HR}}, I^u_{\mathrm{LR}}, R_{\mathrm{HR}}, R_{\mathrm{LR}}, N\right) \tag{3}$$

where $\Psi(\cdot)$ denotes the image sequence construction function, R_{HR} and R_{LR} denote the resolutions of LR and HR images, respectively, and N indicates the total number of images in the sequence. I^s_n is the nth pseudo-image in the generated sequence, and R_n is the corresponding pseudo-resolution.

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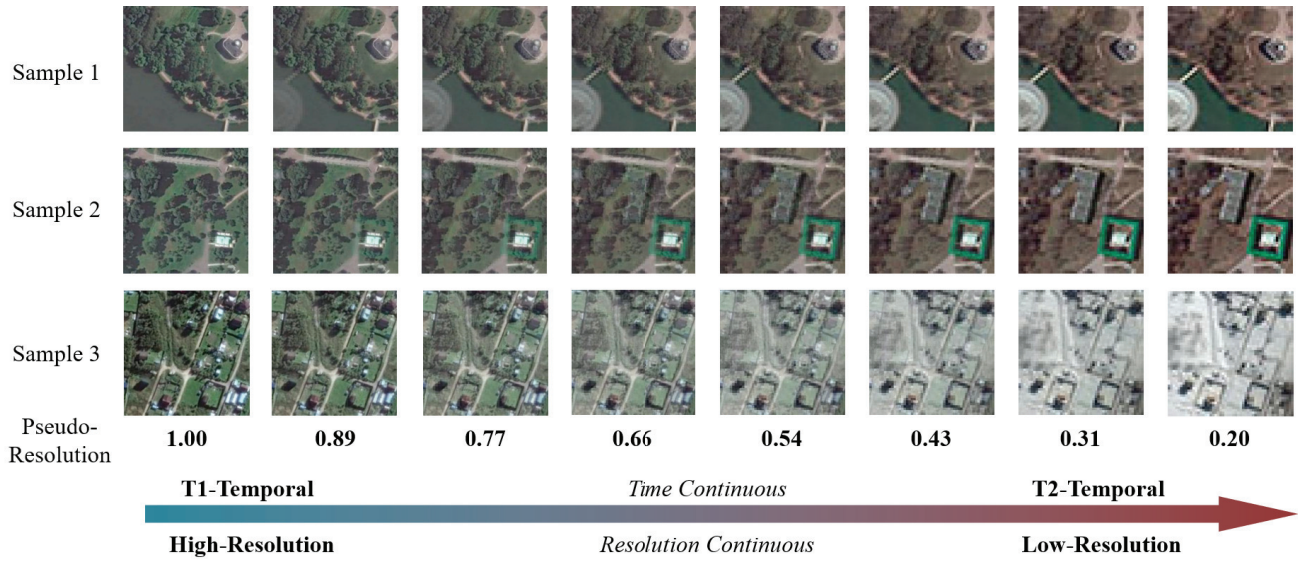


Fig. 4. Constructed pseudo-image sequence with continuous resolution transition. (Some visualization samples from the CDD dataset with a fivefold resolution difference between T1- and T2-temporal images, and the pseudo-resolution of each image in the sequence is expressed as a ratio relative to the HR baseline at the T1-temporal image.)

Based on the constructed pseudo-image sequence with continuous resolution transition, the image sequence encoder extracts the spatiotemporal features of the changed landcovers. The encoder consists of a feature fusion module and three stacked convolutional layers. The feature fusion module uses the resolution sequence corresponding to the image sequence to guide the weighting of multitemporal image features. The convolutional layers further process the fused features to generate higher level semantic features. The output of the image sequence encoder can be represented as S_0–S_3. The principles and details of the PRSR method are detailed in Section III-B.

The bitemporal encoder employs a pretrained ResNet18 [40] as a Siamese network to capture fundamental feature representations from bitemporal images, as illustrated below

$$\begin{aligned} f_{10}, f_{11}, f_{12}, f_{13} &= \Phi_{\text{Siamese}}\left(I_{\text{HR}}\right) \\ f_{20}, f_{21}, f_{22}, f_{23} &= \Phi_{\text{Siamese}}\left(I_{\text{LR}}^{u}\right). \end{aligned} \tag{4}$$

However, the Siamese network inherently lacks multiresolution adaptive feature extraction capability, rendering the acquired features insufficient to effectively represent changed landcovers. To address this limitation, RDENet incorporates a BRDM mechanism within the bitemporal encoder, where the resolution-difference factor r is embedded as a guiding cue. This design enables adaptive enhancement of bitemporal cross-resolution key features, thereby significantly improving the model's capability to detect changed landcovers across cross-resolution scenarios. The aforementioned process can be formally expressed as follows:

$$\begin{aligned} &F_0, F_1, F_2, F_3 \\ &\quad = \text{BRDM}\left(\{f_{10}, f_{11}, f_{12}, f_{13}\}, \{f_{20}, f_{21}, f_{22}, f_{23}\}, r\right) \end{aligned} \tag{5}$$

where BRDM(·) represents the BRDM function and F_0–F_3 represent the enhanced features. F_0–F_3 are subsequently fed into the decoder as the output of the bitemporal encoder, participating in the generation of CD results. The principle and details of the BRDM method are detailed in Section III-C.

The outputs S_0–S_3 from the image sequence encoder and F_0–F_3 from the bitemporal encoder are fed into the decoder. The decoder progressively restores detailed information through operations including channel concatenation and upsampling, ultimately generating the CD predictions.

B. Pseudo-Continuous Resolution Sequence Representation

PRSR constructs a pseudo-image sequence from the prechanged image to the postchanged image, and this method was inspired by Lin et al. [41]. The sequence shows continuous transitions in both time and resolution dimensions. The adjacent pseudo-images in the sequence have small differences in spatial details. Compared with the direct comparison of HR and LR images, the pseudo-image sequence enables the model to gradually adapt to multiresolution features through a gradual resolution transition, thereby mitigating interference caused by resolution abruptness. Furthermore, we propose a RGFF module that enhances the model's capability to capture spatial features of changed landcovers, ultimately achieving accurate cross-resolution CD.Specifically, a pseudo-image sequence is constructed using linear interpolation, spanning from the prechange image (T1) to the postchange image (T2), thereby generating a continuous pseudo-image sequence composed of N images, so the image sequence construction function in (3) can be detailed as

$$\begin{aligned} I_n^s &= \left(1 - \frac{n}{N-1}\right) \times I_{\text{HR}} + \frac{n}{N-1} \times I_{\text{LR}}^{u} \\ R_n &= \left(1 - \frac{n}{N-1}\right) \times R_{\text{HR}} + \frac{n}{N-1} \times R_{\text{LR}} \end{aligned} \tag{6}$$

where I_{HR} and I_{LR}^{u} represent the HR and upsampled LR images, respectively, R_{HR} and R_{LR} denote their corresponding resolutions, and N indicates the number of images contained in the sequence. Fig. 4 provides visualization examples of the

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constructed pseudo-image sequence. The pseudo-resolution sequence R_1–R_N serves as prior resolution information for each image, and in subsequent computations, feature weights are adaptively generated to perform weighted fusion of features at different resolutions.

In the constructed pseudo-image sequence, the changed regions exhibit a gradual evolutionary process. Correspondingly, an image sequence encoder is designed to extract spatiotemporal features from continuous-resolution image sequences. To reduce parameter storage and computational overhead, each encoding stage employs 3×3 and 1×1 convolutional layers for feature extraction. Building upon this architecture, a RGFF module is implemented to perform adaptive weighted fusion of features at different resolutions, ultimately achieving effective spatiotemporal feature extraction. Specifically, the encoder adopts a four-stage architecture, with each stage executing the following computational process:

$$F_{i+1,n} = \text{Pool}\left(\text{BN}\left(\text{Conv}_{1\times1}\left(\text{BN}\left(\text{Conv}_{3\times3}\left(F_{i,n}\right)\right)\right)\right)\right) \quad (7)$$

where i and $i+1$ denote the ith and $(i+1)$th stages in the image sequence encoder, n represents the feature corresponding to the nth image in the pseudo-image sequence, Pool$(\cdot)$ denotes the pooling operation, and BN$(\cdot)$ represents the batch normalization operation.

On this basis, the RGFF module is performed on features of different resolutions in each stage to achieve spatiotemporal feature extraction, which is beneficial for cross-resolution changed landcover identification. In contrast to conventional feature fusion methods, the proposed module employs the pseudo-resolution of each feature's corresponding image as guidance to generate adaptive weights for weighted fusion of multiresolution features, as follows:

$$S_i = \text{Conv}_{3\times3}\left(\sum_{n=1}^{N}\left\{F_{i,n} \times \text{Conv}_{1\times1}\left(R_n\right)\right\}\right) \quad (8)$$

where $F_{i,n}$ represents the feature of the nth image at the ith encoding stage, while R_n denotes the corresponding pseudo-resolution generated by (6).

The architectures of the image sequence encoder and RGFF module are shown in Fig. 5. The outputs S_0–S_3 of the RGFF module characterize the spatiotemporal features of changed landcovers in a cross-resolution pseudo-image sequence, effectively mitigating the challenges in extracting change features caused by resolution discrepancies between prechange and postchange images. These features are subsequently fed into the change decoder to participate in the generation of CD results.

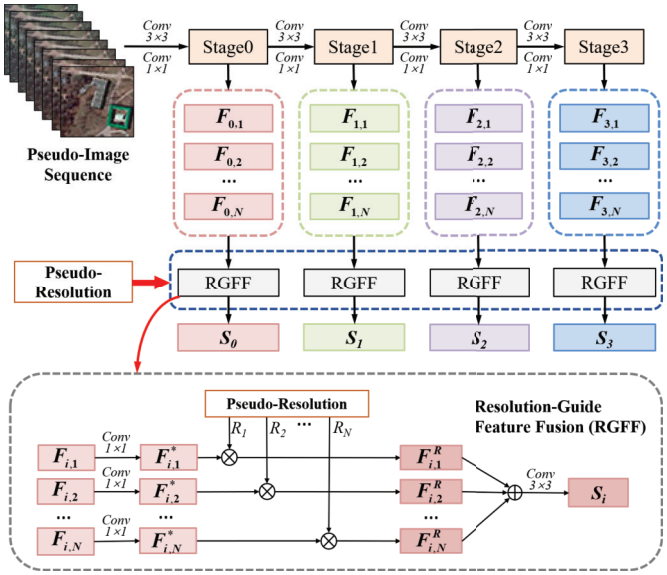


Fig. 5. Schematic of image sequence encoder and RGFF module. The input to the image sequence encoder is the pseudo-image sequence, and features are extracted through four encoding stages. The features output from each stage are fed into the RGFF module for adaptive weighted fusion.

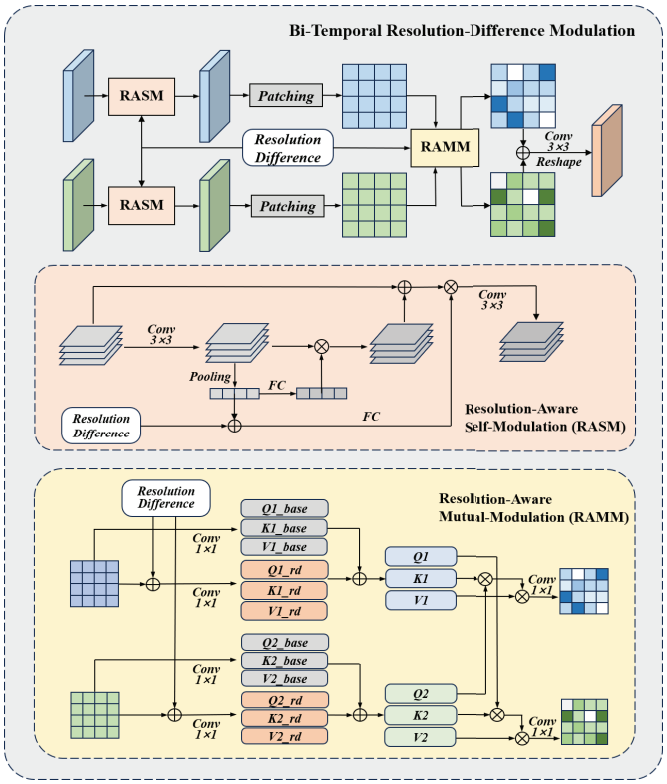


Fig. 6. Schematic of the BRDM method. The BRDM includes two components: RASM and RAMM. RASM adaptively enhances features in the bitemporal images, while RAMM enables cross-temporal feature interaction.

C. Bitemporal Resolution-Difference Modulation

The conventional Siamese network struggles to simultaneously characterize features of images with varying resolutions, which makes it difficult to distinguish the changed and unchanged landcovers. Therefore, to enhance the model's perception capability to the key features of the landcovers and distinguish between the changed and unchanged landcovers, a BRDM method is proposed. In this method, the resolution-difference factor is used as the prior information to weight the key features of the changed landcovers in cross-resolution image pairs. It consists of two components: RASM and RAMM, with its specific architecture depicted in Fig. 6. RASM performs differentiated enhancement on features

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in bitemporal images, while RAMM enables cross-temporal feature interaction to extract change information.

The RASM adaptively weights the features of bitemporal images by embedding the resolution-difference factor into channelwise features to generate adaptive feature weights. Specifically, convolution and pooling operations are employed to capture channelwise features of the feature map

$$f_c = \text{Pool}\left(\text{Conv}_{3\times3}\left(F_{in}\right)\right). \tag{9}$$

The expanded resolution-difference factor is first elementwise added with the channelwise feature, followed by processing through a fully connected layer to obtain resolution-aware feature weight

$$v = \text{FC}\left(f_c + \text{Expd}\left(r\right)\right) \tag{10}$$

where FC(·) denotes the fully connected layer; Expd(·) indicates the expansion operation to align the resolution-difference factor with the channelwise feature in dimensionality.The input feature map is first adaptively weighted by channelwise feature, and then further weighted by the resolution-embedded weight obtained from (10), which can be expressed as

$$\begin{aligned} F_{\text{CF}} &= F + \text{Conv}_{3\times3}\left(F\right) \times \text{FC}\left(f_c\right) \\ F_S &= \text{Conv}_{3\times3}\left(v \times F_{\text{CF}}\right) \end{aligned} \tag{11}$$

where F denotes the input feature map to the RASM module and F_S denotes the output feature map.

The RASM module enhances the model's ability to extract features from images with different resolutions by embedding resolution difference into the feature extraction process. Building on this, a RAMM module is introduced to facilitate interactive enhancement of cross-resolution bitemporal features.

The RAMM employs a query–key–value (Q–K–V) mechanism for cross-resolution feature interaction. Unlike conventional self-attention's Q–K–V computation [30], the proposed method embeds resolution difference into the learning process. Specifically, the Q, K, and V vectors of bitemporal features are generated through two components, the first part is obtained by partitioning and projecting the original feature maps

$$\begin{aligned} Q_{\text{base}} &= \text{Conv}_{1\times1}^{\text{Qbase}}\left(\text{Patch}\left(F_S\right)\right) \\ K_{\text{base}} &= \text{Conv}_{1\times1}^{\text{Kbase}}\left(\text{Patch}\left(F_S\right)\right) \\ V_{\text{base}} &= \text{Conv}_{1\times1}^{\text{Vbase}}\left(\text{Patch}\left(F_S\right)\right) \end{aligned} \tag{12}$$

where $\text{Conv}_{1\times1}^{\text{Qbase}}$, $\text{Conv}_{1\times1}^{\text{Kbase}}$, and $\text{Conv}_{1\times1}^{\text{Vbase}}$ denote 1×1 convolution operations used to generate the Q_{base}, K_{base}, and V_{base} vectors.

While the second part is obtained by first partitioning the original features, then integrating the resolution-difference factor through elementwise addition, and finally projecting the fused representations via a convolutional layer

$$\begin{aligned} Q_{\text{rd}} &= \text{Conv}_{1\times1}^{\text{Qrd}}\left(\text{Patch}\left(F_S\right) + \text{Expd}\left(r\right)\right) \\ K_{\text{rd}} &= \text{Conv}_{1\times1}^{\text{Krd}}\left(\text{Patch}\left(F_S\right) + \text{Expd}\left(r\right)\right) \\ V_{\text{rd}} &= \text{Conv}_{1\times1}^{\text{Vrd}}\left(\text{Patch}\left(F_S\right) + \text{Expd}\left(r\right)\right) \end{aligned} \tag{13}$$

where $\text{Conv}_{1\times1}^{\text{Qrd}}$, $\text{Conv}_{1\times1}^{\text{Krd}}$, and $\text{Conv}_{1\times1}^{\text{Vrd}}$ denote 1×1 convolution operations used to generate the Q_{rd}, K_{rd} and V_{rd} vectors.

The Q, K, and V are formed by the sum of these two parts

$$\{Q, K, V\} = \{Q_{\text{base}}, K_{\text{base}}, V_{\text{base}}\} + \{Q_{\text{rd}}, K_{\text{rd}}, V_{\text{rd}}\}. \tag{14}$$

The cross-temporal Q and K vectors establish interfeature correlations across different temporal images. The similarity scores are computed via dot product operations to determine the weighting coefficients for each V. This process can be formally expressed as

$$\begin{aligned} \Gamma\left(Q_2, K_1, V_1\right) &= \text{softmax}\left(\frac{Q_2 K_1^{\text{T}}}{\sqrt{d_{k1}}}\right) \cdot V_1 \\ \Gamma\left(Q_1, K_2, V_2\right) &= \text{softmax}\left(\frac{Q_1 K_2^{\text{T}}}{\sqrt{d_{k2}}}\right) \cdot V_2 \end{aligned} \tag{15}$$

where the subscripts 1 and 2 denote the first and second temporal phases, respectively, and d_{k1} and d_{k2} denote the dimensionalities of the K_1 and K_2 vectors.

The outputs from the aforementioned computation process undergo 3×3 convolution to generate mutually modulated enhanced features F_{M1} and F_{M2}. These features are then processed through elementwise addition, a convolutional layer, and a reshape operation, ultimately producing the final output

$$F_{\text{OUT}} = \text{Reshape}\left(\text{Conv}_{3\times3}\left(F_{M1} + F_{M2}\right)\right). \tag{16}$$

The outputs corresponding to encoding stages 0–3 are denoted as F_0–F_3. These features F_0–F_3, serving as inputs to the subsequent change decoder, collaboratively generate the final CD results with the outputs S_0–S_3 from the image sequence encoder.

D. Loss Function

As shown in Fig. 3, in addition to the final prediction output of the decoder, a convolutional layer and an upsampling layer are added after each stage of the decoder's output, producing auxiliary prediction results that match the size of the ground truth. These auxiliary prediction results are used to compute auxiliary losses in a deeply supervised manner. The auxiliary losses can alleviate the vanishing gradient problem [33], [43], accelerate the convergence of deep models, and directly apply feedback from the ground truth to the hidden layers of the model, enabling the model to learn more discriminative feature representations and further improve CD performance [29]. The total loss function can be expressed as follows:

$$\begin{aligned} \mathcal{L}_{\text{total}} &= \mathcal{L}_{\text{pre}} + \mathcal{L}_{\text{aux}} \\ &= \mathcal{L}_{\text{ce}}\left(P, G\right) + \sum_{i=1}^{S} \mathcal{L}_{\text{ce}}\left(P_i, G\right) \end{aligned} \tag{17}$$

where $\mathcal{L}_{\text{pre}}$ represents the prediction loss, $\mathcal{L}_{\text{aux}}$ represents the auxiliary prediction loss, $\mathcal{L}_{\text{ce}}$ is the cross-entropy loss, P is the CD prediction output by the decoder, P_i is the prediction for ith stage in the total of S stages in decoder, and G represents the ground truth.

Based on this, the model parameters can be updated according to the loss function at each training iteration. During training, RDENet employs the standard adaptive moment estimation (Adam) optimizer to update the model parameters until stable convergence is achieved. Based on the above, Algorithm 1 provides a detailed description of the computation and training process of RDENet.

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Algorithm 1 Computation and Training Process of the Proposed RDENet

Require:
The bi-temporal cross-resolution images I_{HR} and I_{LR}, and the corresponding ground truth G;

Ensure:
The RDENet model's parameters Θ;

Randomly initialize the parameters Θ;
Step 1: Image Sequence Encoder
Construct the pseudo-image sequence: $\{I_1^s, I_2^s, \ldots, I_N^s\}$;
Three stacked convolutional layers and RGFF modules for extracting features S_0, S_1, S_2, S_3;
Step 2: Bi-Temporal Encoder
Siamese network extracts features $\{f_{10}, f_{11}, f_{12}, f_{13}\}$ and $\{f_{20}, f_{21}, f_{22}, f_{23}\}$;
Adaptive feature enhancement via the BRDM method: $F_0, F_1, F_2, F_3 = BRDM(\{f_{10}, f_{11}, f_{12}, f_{13}\}, \{f_{20}, f_{21}, f_{22}, f_{23}\})$;
Step 3: Change Decoder
By integrating the outputs of the image sequence encoder and the bi-temporal encoder, the CD results are generated.
Step 4: Back Propagation
Calculate the prediction loss $\mathcal{L}_{pre}$;
Calculate the auxiliary prediction loss $\mathcal{L}_{aux}$;
$\Theta \leftarrow argmin(\mathcal{L}_{pre} + \mathcal{L}_{aux})$;
return: Θ;

IV. Experiments

A. Experimental Settings

1) Datasets Description: The effectiveness of the proposed method is validated through experiments conducted on three publicly available remote sensing CD datasets: CDD [44], WHU-CD [45], and LEVIR-CD [46]. This multidataset evaluation framework ensures comprehensive performance assessment across diverse scenarios.

The CDD dataset [44] comprises 16 000 remote sensing image pairs with dimensions of 256 × 256 pixels, capturing various change types induced by vehicles, roads, buildings, and other urban elements. The dataset is partitioned into training, validation, and testing sets containing 10 000, 3000, and 3000 image pairs, respectively. To simulate cross-resolution scenarios, we perform 7× downsampling on each postchange image (T2), resulting in the cross-resolution dataset CDD(7×) where T2 images exhibit spatial resolution five times lower than their corresponding T1 counterparts.

The WHU-CD dataset [45] consists of two remote sensing images with dimensions of 32 507 × 15 354 pixels, documenting postearthquake building reconstruction. Following the methodology in [30], these images are systematically cropped into 7434 nonoverlapping 256 × 256 pixel image pairs, which are randomly divided into training, validation, and testing sets following an 8:1:1 ratio. For each image pair, we perform 6× downsampling on the prechange image (T1) to obtain the cross-resolution WHU-CD(6×).

The LEVIR-CD dataset [46] serves as a building CD benchmark containing 637 pairs of bitemporal images, each originally sized at 1024 × 1024 pixels. Following [10], each image pair is cropped into nonoverlapping 256 × 256 pixel patches, and these patch pairs are randomly divided into training, validation, and testing sets with a 7:2:1 ratio. For each image pair, we perform 4× downsampling on the postchange image (T2) to obtain the cross-resolution LEVIR-CD(4×).

2) Implementation Details: All experiments were implemented on an NVIDIA Quadro RTX 8000 GPU with the PyTorch framework. For data augmentation, we employed random horizontal flipping, random vertical flipping, and random rotation. The number of training epochs was set to 100 with a batch size of 8, using Adam as the optimizer. The learning rate was initialized to 0.0001 and halved every eight epochs. The number of images in the pseudo-image sequence was set to 8. The Siamese network in the bitemporal encoder is the pretrained ResNet18.

3) Evaluation Metrics: To comprehensively compare the CD accuracy of the proposed RDENet with existing methods, four quantitative metrics were adopted: the F1-score (F1), mean intersection over union (mIoU), Kappa coefficient (Kappa), and overall accuracy (OA). The calculation of these metrics is as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{18}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{19}$$

$$\text{PE} = \frac{(\text{TP} + \text{FN}) \times (\text{TP} + \text{FP})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2} + \frac{(\text{TN} + \text{FP}) \times (\text{TN} + \text{FN})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2} \tag{20}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{21}$$

$$F1 = \frac{2}{\text{Precision}^{-1} + \text{Recall}^{-1}} \tag{22}$$

$$\text{mIoU} = \left(\frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} + \frac{\text{TN}}{\text{TN} + \text{FN} + \text{FP}} \right) \Big/ 2 \tag{23}$$

$$\text{Kappa} = \frac{\text{OA} - \text{PE}}{1 - \text{PE}}. \tag{24}$$

In addition, the number of model parameters (denoted as Param) was utilized to compare the parameter counts across different methods.

B. Experimental Results

1) Comparison With Other Methods: Several recently proposed state-of-the-art CD methods were compared with the proposed RDENet, including: three fully convolutional methods (FC-EF [19], FC-Siam-diff [19], and FC-Siam-conc [19]), two attention-based methods (DSIFN [29] and DMFANet [47]), and two Transformer-based methods (ChangeFormer [24] and BIT [23]). In addition, four CD methods designed for cross-resolution scenarios (SUNet [37], SRCDNet [9], SILI [10], and DCILNet [48]) were also included in the comparison.

The proposed RDENet and the aforementioned CD methods were applied to three widely used remote sensing CD datasets: CDD(7×), WHU-CD(6×), and LEVIR-CD(4×). Notably, to adapt conventional CD methods to cross-resolution CD tasks,

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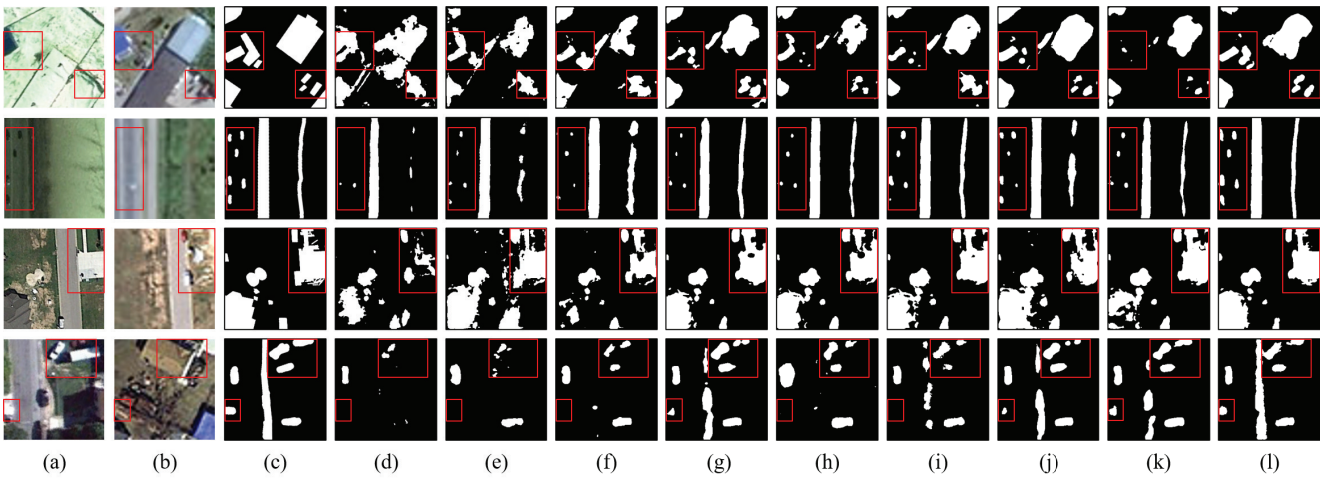


Fig. 7. Visual comparisons among the CD results produced by different methods on cross-resolution CDD(7×) dataset. (a) T1-temporal image (HR). (b) T2-temporal image (LR). (c) Ground truth. (d) DSIFN. (e) ChangeFormer. (f) BIT. (g) DMFANet. (h) SUNet. (i) SRCDNet. (j) SILI. (k) DCILNet. (l) Proposed RDENet. The changed areas are in white, while the unchanged areas are in black.

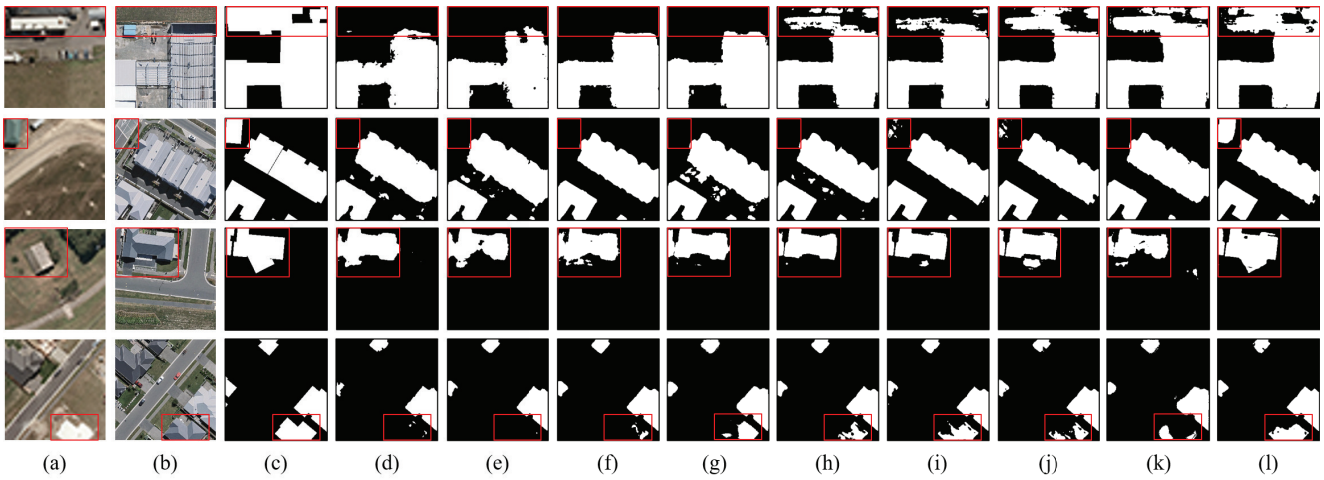


Fig. 8. Visual comparisons among the CD results produced by different methods on cross-resolution WHU-CD(6×) dataset. (a) T1-temporal image (LR). (b) T2-temporal image (HR). (c) Ground truth. (d) DSIFN. (e) ChangeFormer. (f) BIT. (g) DMFANet. (h) SUNet. (i) SRCDNet. (j) SILI. (k) DCILNet. (l) Proposed RDENet. The changed areas are in white, while the unchanged areas are in black.

LR images were resized to match the dimensions of HR images via bicubic interpolation before being fed into the CD models. Quantitative comparison results are summarized in Tables I–III. Furthermore, visual comparisons were conducted to validate the aforementioned results, as illustrated in Figs. 7–9.

The experimental results for CDD(7×) are presented in Table I. The findings demonstrate that the proposed RDENet achieves outstanding accuracy on the CDD(7×) dataset. Specifically, RDENet attains remarkable evaluation metrics of 95.38% F1-score, 94.99% mIoU, 94.78% Kappa, and 98.94% OA. The CDD(7×) dataset contains numerous remote sensing image pairs with significant seasonal variations, posing challenges for identifying changed landcovers like vehicles, roads, and buildings in cross-resolution scenarios. Furthermore, interference factors such as illumination changes and shadow effects may lead to false detections. Despite these challenges, the proposed RDENet demonstrates satisfactory performance compared with other methods. As shown in Fig. 7 through visual comparisons on the CDD(7×) dataset, RDENet exhibits substantial improvements in changed landcover extraction under complex cross-resolution scenarios involving seasonal differences, illumination variations, and shadow interference.

TABLE I
QUANTITATIVE COMPARISON WITH OTHER METHODS ON CROSS-RESOLUTION CDD(7×) DATASET

Method	F1(%)	mIoU(%)	Kappa(%)	OA(%)
FC-EF [19]	80.23	81.13	77.82	95.69
FC-Siam-diff [19]	81.27	81.89	78.83	95.76
FC-Siam-conc [19]	82.46	82.93	80.27	96.10
DSIFN [29]	86.17	86.11	84.40	96.86
ChangeFormer [24]	90.17	89.80	88.91	97.76
BIT [23]	89.88	89.52	88.57	97.68
DMFANet [47]	93.43	92.99	92.58	98.50
SUNet [37]	92.05	91.61	91.02	98.17
SRCDNet [9]	93.66	93.23	92.84	98.55
SILI [10]	94.01	93.58	93.24	98.63
DCILNet [48]	93.74	93.30	92.92	98.55
RDENet(proposed)	**95.38**	**94.99**	**94.78**	**98.94**

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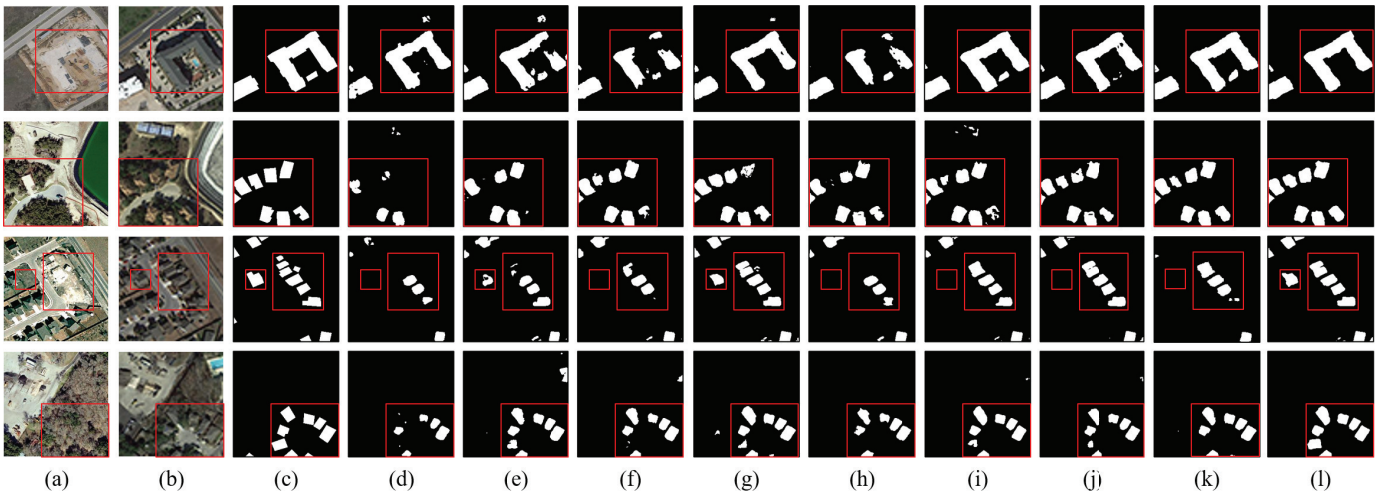


Fig. 9. Visual comparisons among the CD results produced by different methods on cross-resolution LEVIR-CD(4×) dataset. (a) T1-temporal image (HR). (b) T2-temporal image (LR). (c) Ground truth. (d) DSIFN. (e) ChangeFormer. (f) BIT. (g) DMFANet. (h) SUNet. (i) SRCDNet. (j) SILI. (k) DCILNet. (l) Proposed RDENet. The changed areas are in white, while the unchanged areas are in black.

TABLE II
QUANTITATIVE COMPARISON WITH OTHER METHODS ON CROSS-RESOLUTION WHU-CD(6×) DATASET

Method	F1(%)	mIoU(%)	Kappa(%)	OA(%)
FC-EF [19]	54.36	65.94	51.62	94.72
FC-Siam-diff [19]	59.18	68.41	56.62	94.98
FC-Siam-conc [19]	60.45	69.42	58.18	95.67
DSIFN [29]	88.55	89.13	87.95	98.87
ChangeFormer [24]	88.34	88.95	87.73	98.84
BIT [23]	89.69	90.11	89.15	98.97
DMFANet [47]	92.31	92.45	91.90	99.23
SUNet [37]	91.13	91.37	90.65	99.08
SRCDNet [9]	91.96	92.12	91.52	99.17
SILI [10]	92.18	92.32	91.76	99.19
DCILNet [48]	92.68	92.79	92.29	99.26
RDENet(proposed)	**93.48**	**93.53**	**93.13**	**99.33**

TABLE III
QUANTITATIVE COMPARISON WITH OTHER METHODS ON CROSS-RESOLUTION LEVIR-CD(4×) DATASET

Method	F1(%)	mIoU(%)	Kappa(%)	OA(%)
FC-EF [19]	74.38	78.35	73.12	97.58
FC-Siam-diff [19]	75.76	79.29	74.55	97.69
FC-Siam-conc [19]	72.44	77.07	71.12	97.45
DSIFN [29]	84.57	85.84	83.77	98.48
ChangeFormer [24]	85.32	86.45	84.56	98.56
BIT [23]	86.37	87.30	85.66	98.65
DMFANet [47]	92.31	92.45	91.90	99.23
SUNet [37]	85.11	86.27	84.34	98.53
SRCDNet [9]	84.76	86.01	83.99	98.52
SILI [10]	88.43	89.02	87.82	98.84
DCILNet [48]	92.68	92.79	92.29	99.26
RDENet(proposed)	**91.27**	**91.51**	**90.81**	**99.12**

TABLE IV
MODEL EFFICIENCY COMPARISON WITH OTHER METHODS

Method	Params(M)	FLOPs(G)	Inference Time(ms)
FC-EF [19]	1.35	7.14	6.89
FC-Siam-diff [19]	1.35	9.45	7.53
FC-Siam-conc [19]	1.55	10.62	7.66
DSIFN [29]	35.73	164.53	19.45
ChangeFormer [24]	24.30	16.84	8.83
BIT [23]	3.01	17.00	8.20
DMFANet [47]	7.18	38.48	9.16
SUNet [37]	15.56	79.77	10.10
SRCDNet [9]	17.00	147.95	16.30
SILI [10]	13.06	17.50	8.27
DCILNet [48]	46.86	21.36	9.03
RDENet(proposed)	10.93	29.58	9.09

The experimental results on the WHU-CD(6×) dataset are shown in Table II. The WHU-CD(6×) dataset is primarily used to assess building changes after seismic disasters. In cross-resolution scenarios, significant differences exist in roof structures and complex textures of buildings between prechange and postchange images, while vehicles and roads introduce additional interference. The experimental results demonstrate that the proposed RDENet outperforms other methods on the WHU-CD(6×) dataset. In terms of quantitative metrics, RDENet achieves detection results of 93.48% F1-score, 93.53% mIoU, 93.13% Kappa, and 99.33% OA. As illustrated in Fig. 8, RDENet demonstrates exceptional performance in building CD under cross-resolution scenarios on the WHU-CD(6×) dataset.

The experimental results on the LEVIR-CD(4×) dataset are presented in Table III. The LEVIR-CD(4×) dataset is designed for the CD of buildings such as residential houses, apartments, and warehouses. In cross-resolution scenarios, these buildings are susceptible to interference from complex surrounding environments, making it challenging to generate accurate extraction results. The experimental results in Table III demonstrate that the proposed RDENet outperforms other methods on the LEVIR-CD(4×) dataset. RDENet achieves outstanding detection performance, with an F1-score of 91.27%, mIoU of 91.51%, Kappa of 90.81%, and OA of 99.12%. As shown in Fig. 9 through visual comparisons, RDENet exhibits superior

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TABLE V
QUANTITATIVE COMPARISON WITH OTHER METHODS UNDER DIFFERENT RESOLUTION-DIFFERENCE CONDITIONS ON CDD DATASET

Method	4×		6×		7×		8×		9×		10×	
	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)
FC-EF [19]	84.16	84.39	83.27	83.59	80.23	81.13	78.68	79.86	75.21	77.16	73.01	75.51
FC-Siam-diff [19]	84.91	85.04	84.76	84.93	81.27	81.89	78.99	80.11	77.02	78.51	73.71	76.04
FC-Siam-conc [19]	85.11	85.21	85.01	85.11	82.46	82.93	80.12	81.02	78.51	79.70	74.02	76.27
DSIFN [29]	89.52	89.16	89.32	88.99	86.17	86.11	83.93	84.15	80.15	80.98	76.98	78.43
ChangeFormer [24]	91.77	91.35	90.99	90.58	90.17	89.80	87.04	86.89	86.17	86.12	84.16	84.34
BIT [23]	92.07	91.64	91.04	90.63	89.88	89.52	88.82	88.52	87.27	87.12	85.37	85.42
DMFANet [47]	93.87	93.43	93.59	93.15	93.43	92.99	93.16	92.72	92.87	92.42	92.02	91.58
SUNet [37]	92.68	92.24	92.60	92.16	92.05	91.61	91.80	91.39	89.95	89.60	87.72	87.51
SRCDNet [9]	94.01	93.58	93.71	93.27	93.66	93.23	93.14	92.70	93.01	92.58	91.04	90.64
SILI [10]	94.19	93.77	94.10	93.67	94.01	93.58	93.87	93.44	93.16	92.73	91.87	91.44
DCILNet [48]	94.12	93.69	93.97	93.54	93.74	93.30	93.41	92.97	93.12	92.67	92.41	91.97
RDENet(proposed)	**95.96**	**95.60**	**95.88**	**95.51**	**95.38**	**94.99**	**94.96**	**94.56**	**94.38**	**93.96**	**93.88**	**93.45**

TABLE VI
QUANTITATIVE COMPARISON WITH OTHER METHODS UNDER DIFFERENT RESOLUTION-DIFFERENCE CONDITIONS ON WHU-CD DATASET

Method	4×		6×		7×		8×		9×		10×	
	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)
FC-EF [19]	54.97	66.49	54.36	65.94	54.11	66.52	51.98	65.49	51.29	65.16	50.11	64.58
FC-Siam-diff [19]	61.21	70.03	59.18	68.41	58.64	69.00	57.97	68.60	57.01	68.10	55.37	67.24
FC-Siam-conc [19]	61.78	70.35	60.45	69.42	60.08	69.77	58.32	68.78	57.68	68.45	56.02	67.61
DSIFN [29]	88.64	89.18	88.55	89.10	86.98	87.93	85.59	86.80	84.01	85.54	83.06	84.80
ChangeFormer [24]	89.32	89.77	88.34	88.38	87.21	88.13	86.28	87.36	85.59	86.79	83.74	85.33
BIT [23]	90.16	90.52	89.69	89.76	87.79	88.61	86.04	87.16	85.71	86.89	84.12	85.63
DMFANet [47]	92.65	92.77	92.31	92.45	91.83	92.01	90.42	90.75	88.79	89.33	88.19	88.82
SUNet [37]	91.62	91.83	91.13	91.40	90.04	90.52	88.46	89.17	86.62	87.63	84.78	86.15
SRCDNet [9]	92.14	92.30	91.96	92.44	91.02	91.38	89.17	89.77	88.10	88.87	87.01	87.96
SILI [10]	92.57	92.70	92.18	92.40	91.17	91.52	90.04	90.52	88.59	89.28	88.13	88.89
DCILNet [48]	93.01	93.10	92.68	92.79	92.05	92.22	90.76	91.05	89.02	89.53	88.36	88.97
RDENet(proposed)	**93.62**	**93.67**	**93.48**	**93.53**	**92.34**	**92.56**	**91.02**	**91.38**	**90.64**	**91.05**	**89.24**	**89.83**

performance in detecting multitype building changes under cross-resolution scenarios compared with other methods.

We quantified the number of parameters (Params), floating-point operations (FLOPs), and inference time for all methods, as presented in Table IV. The results indicate that comparative models like DSIFN (35.73M parameters) and DCILNet (46.86M parameters) require significantly larger parameter counts, while SRCDNet (147.95G FLOPs) demonstrates excessively high computational demands, leading to elevated computational costs. Although FC-EF, FC-Siam-diff, FC-Siam-conc, and BIT are lightweight CD models, their limited detection accuracy fails to meet practical requirements. The proposed RDENet (10.93M parameters and 29.58G FLOPs) achieves a balanced tradeoff between computational efficiency and detection accuracy. While the introduced image sequence encoder in RDENet increases computational throughput, the image sequence feature fusion significantly enhances CD accuracy, which is extremely important for the current cross-resolution CD task.

To further validate the detection capability of the model under varying resolution disparities, image pair samples from three datasets with distinct resolution differences were fed into the proposed RDENet, which was trained under the specified resolution-difference ratio settings. The selected resolution difference increased from 4× to 10×. The experimental results are documented in Tables V–VII.

The quantitative results in the tables demonstrate that the proposed RDENet not only consistently outperforms other methods in terms of F1/mIoU scores across three cross-resolution datasets but also exhibits significant performance advantages under larger resolution-difference settings. As evidenced by the results, existing methods achieve relatively high detection accuracy under minor resolution-difference settings, but exhibit significant performance degradation with increasing resolution difference. This phenomenon arises because expanding resolution gaps amplify spatial detail discrepancies in bitemporal images, substantially complicating changed landcover feature extraction. In contrast, the proposed RDENet

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TABLE VII
QUANTITATIVE COMPARISON WITH OTHER METHODS UNDER DIFFERENT RESOLUTION-DIFFERENCE CONDITIONS ON LEVIR-CD DATASET

Method	4×		6×		7×		8×		9×		10×	
	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)
FC-EF [19]	74.38	78.35	73.36	77.60	72.16	76.83	71.90	76.62	68.02	74.15	66.32	73.08
FC-Siam-diff [19]	75.76	79.29	72.69	77.18	71.94	76.68	69.97	75.35	69.13	74.90	67.16	73.58
FC-Siam-conc [19]	72.44	77.07	71.97	76.73	71.21	76.20	70.36	75.65	68.87	74.72	66.39	73.16
DSIFN [29]	84.57	85.85	84.06	85.43	83.57	85.03	83.06	84.64	81.11	83.13	80.89	82.95
ChangeFormer [24]	85.32	86.45	85.16	86.32	83.96	85.36	82.27	84.01	80.92	82.98	78.90	81.47
BIT [23]	86.37	87.30	85.67	86.72	84.86	86.06	83.01	84.59	81.37	83.35	79.79	82.13
DMFANet [47]	90.21	90.56	89.58	90.01	88.07	88.71	86.53	87.42	85.93	86.92	85.14	86.28
SUNet [37]	85.11	86.27	84.03	85.40	82.91	84.52	82.38	84.11	81.10	83.13	80.36	82.57
SRCDNet [9]	84.76	86.01	84.39	85.67	83.69	85.13	82.49	84.19	82.09	83.88	81.13	83.15
SILI [10]	88.43	89.02	87.95	88.61	86.81	87.66	86.13	87.10	85.57	86.63	85.06	86.24
DCILNet [48]	90.68	90.98	89.86	90.26	88.26	88.88	87.19	87.97	86.31	87.23	85.69	86.72
RDENet(proposed)	**91.27**	**91.51**	**90.98**	**91.25**	**90.39**	**90.73**	**89.67**	**90.10**	**88.69**	**89.25**	**88.18**	**88.82**

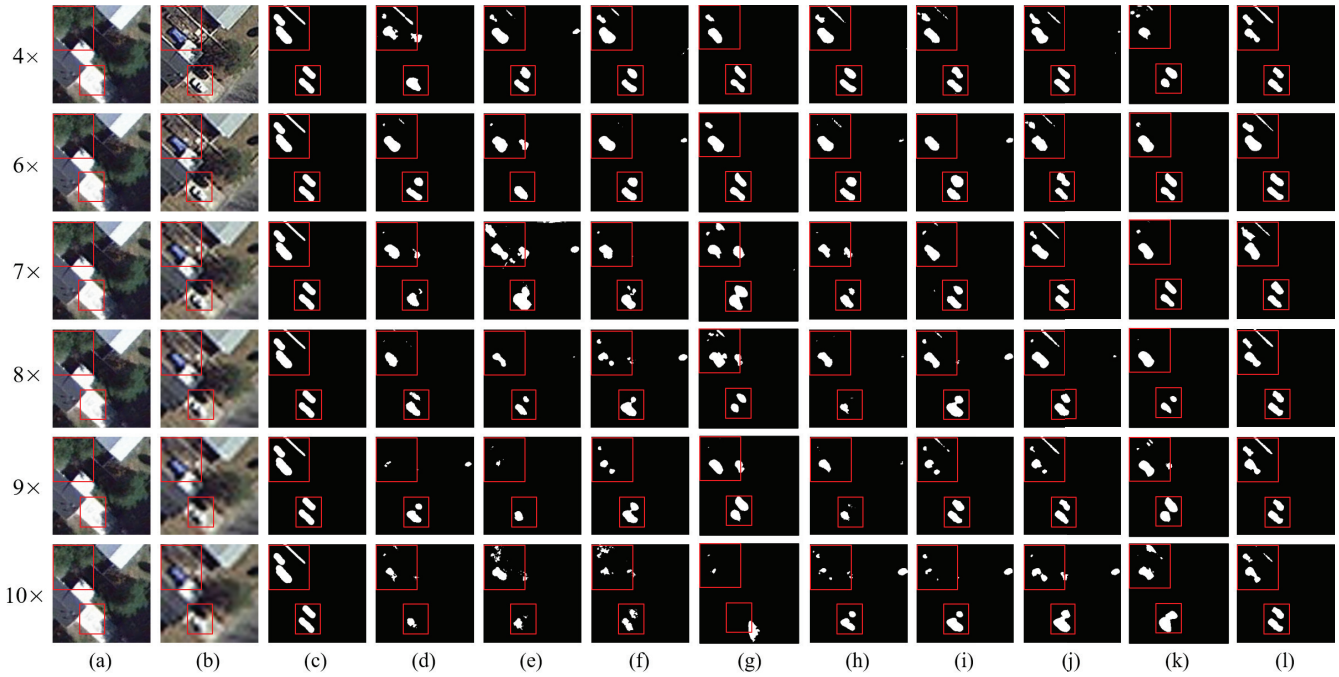


Fig. 10. Visual comparison on the cross-resolution CDD dataset with varying resolution-difference ratios. (a) T1-temporal image (HR). (b) T2-temporal image (LR). (c) Ground truth. (d) DSIFN. (e) ChangeFormer. (f) BIT. (g) DMFANet. (h) SUNet. (i) SRCDNet. (j) SILI. (k) DCILNet. (l) Proposed RDENet.

incorporates resolution difference into its learning paradigm, implicitly capturing changed landcover features across varying resolution conditions, thereby demonstrating superior cross-resolution detection capability.

To intuitively illustrate the adaptability of the proposed RDENet under different resolution differences, some image pair samples of the three datasets are selected, and the detection results of different methods under different resolution differences are visualized, as shown in Figs. 10–12. The main content of Figs. 10–12 focuses on landcover CD in cross-resolution complex scenarios. Under cross-resolution conditions, the loss of spatial details in landcovers and susceptibility to objective factors, such as shadows and illumination, lead to performance degradation in most existing detection methods, which becomes particularly pronounced under significant resolution-difference conditions. The proposed RDENet is minimally affected by resolution differences. As the resolution-difference ratio increases, the detection results exhibit only marginal variations while still substantially outperforming other methods, which illustrates the exceptional adaptability of RDENet in diverse cross-resolution scenarios.

In the aforementioned experiments, we adopted downsampled T2 for CDD/LEVIR-CD and downsampled T1 for WHU-CD to simulate real-world cross-resolution scenarios. The reverse downsampling configuration can effectively validate the effectiveness of the proposed RDENet. Therefore, we

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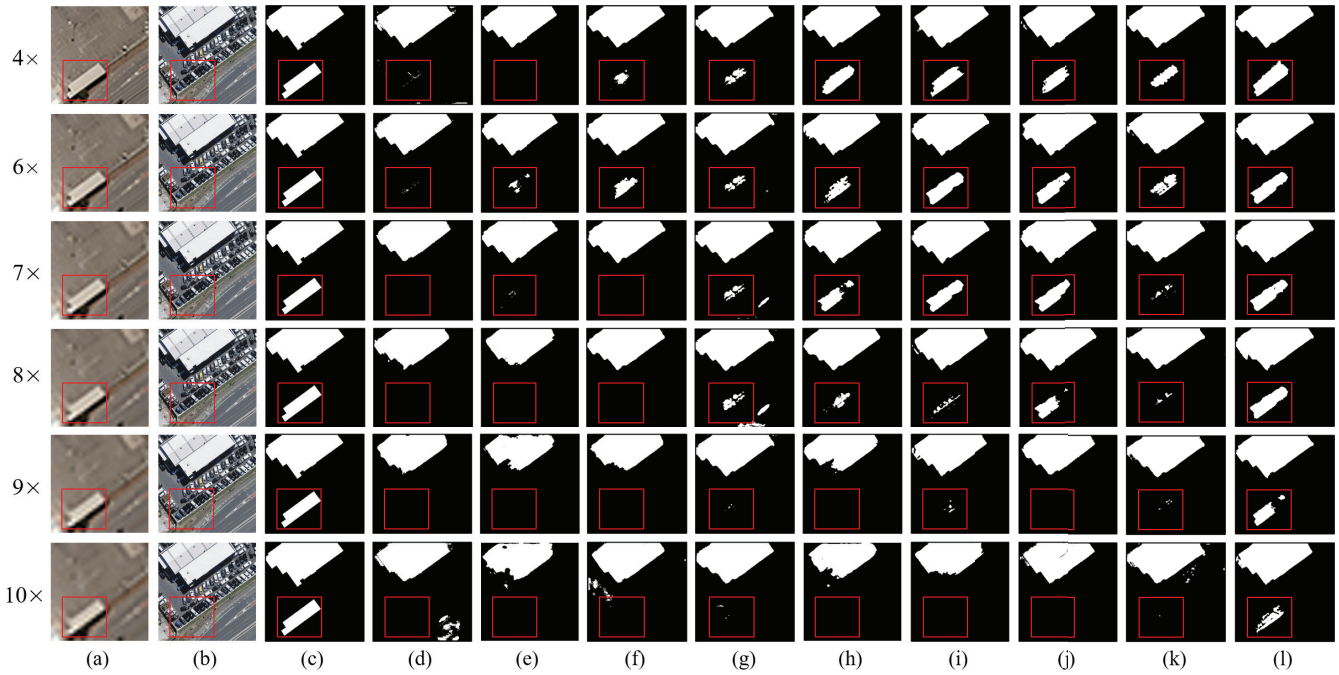


Fig. 11. Visual comparison on the cross-resolution WHU-CD dataset with varying resolution-difference ratios. (a) T1-temporal image (LR). (b) T2-temporal image (HR). (c) Ground truth. (d) DSIFN. (e) ChangeFormer. (f) BIT. (g) DMFANet. (h) SUNet. (i) SRCDNet. (j) SILI. (k) DCILNet. (l) Proposed RDENet.

TABLE VIII

QUANTITATIVE COMPARISON WITH OTHER METHODS UNDER DIFFERENT RESOLUTION-DIFFERENCE CONDITIONS ON CDD DATASET. (ADOPTING THE REVERSED CROSS-RESOLUTION CONFIGURATION, I.E., DOWNSAMPLING T1 WHILE MAINTAINING T2)

Method	4×		6×		7×		8×		9×		10×	
	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)
FC-EF [19]	84.24	84.45	83.68	83.91	81.03	81.72	78.83	79.96	75.73	77.54	73.31	75.72
FC-Siam-diff [19]	84.73	84.88	84.57	84.72	82.07	82.55	78.64	79.79	77.49	78.85	74.11	76.31
FC-Siam-conc [19]	85.42	85.47	85.13	85.20	82.65	83.07	79.98	80.87	77.46	78.88	73.92	76.20
DSIFN [29]	90.01	89.63	89.45	89.11	86.09	86.03	82.57	82.99	81.06	81.70	77.14	78.53
ChangeFormer [24]	91.53	91.10	91.02	90.63	90.47	90.08	87.21	87.02	85.89	85.85	84.59	84.69
BIT [23]	92.27	91.84	90.79	90.39	90.01	89.64	88.35	88.09	86.85	86.71	85.12	85.18
DMFANet [47]	93.65	93.21	93.50	93.06	93.27	92.83	93.09	92.65	92.73	92.29	92.17	91.74
SUNet [37]	92.23	91.80	91.98	91.54	91.64	91.22	91.28	90.88	90.16	89.80	87.91	87.67
SRCDNet [9]	93.79	93.35	93.47	93.03	93.21	92.77	93.10	92.66	92.47	92.02	91.31	90.88
SILI [10]	94.04	93.61	94.01	93.58	93.96	93.53	93.68	93.24	93.33	92.89	91.69	91.27
DCILNet [48]	94.21	93.78	94.02	93.59	93.80	93.36	93.28	92.84	93.05	92.61	92.53	92.09
RDENet(proposed)	**95.73**	**95.35**	**95.47**	**95.08**	**95.38**	**94.99**	**94.73**	**94.32**	**94.27**	**93.84**	**93.67**	**93.23**

added experiments with the reversed downsampling settings, and the results are recorded in Tables VIII–X.

In these additional supplementary experiments, we observed that performance metrics such as F1-score and mIoU generally remained stable or showed minor variations, with no significant degradation. This is attributed to the method's unique design, which mitigates the challenges posed by resolution disparities, making it applicable across various cross-resolution CD scenarios.

2) Comparison Under Equal-Resolution Condition: To more thoroughly validate the effectiveness of the proposed RDENet across various scenarios, we evaluated our method using T1 and T2 images with equal resolution. The results are recorded in Table XI.

Under equal-resolution conditions, as shown in the table, the model demonstrates stronger capability in capturing fine-grained detail features, reducing false positives and missed detections. For instance, on the CDD dataset, the F1-score under equal-resolution input conditions reaches 96.93%, and similar gains were observed on LEVIR-CD and WHU-CD datasets (with F1-scores of 93.94% and 91.79%, respectively). These results highlight the scalability and effectiveness of the proposed method when ideal equal-resolution processing can be achieved.

3) Comparison Under Nonstandard Resolution-Difference Condition: Nonstandard resolution differences at fractional multiples also exist in real-world scenarios. We have added comparative experiments under such nonstandard resolution

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TABLE IX

QUANTITATIVE COMPARISON WITH OTHER METHODS UNDER DIFFERENT RESOLUTION-DIFFERENCE CONDITIONS ON WHU-CD DATASET. (ADOPTING THE REVERSED CROSS-RESOLUTION CONFIGURATION, I.E., DOWNSAMPLING T2 WHILE MAINTAINING T1)

Method	4×		6×		7×		8×		9×		10×	
	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)
FC-EF [19]	55.18	66.58	54.47	66.10	53.83	65.93	51.82	64.95	51.14	64.56	50.24	64.11
FC-Siam-diff [19]	61.27	70.07	59.25	68.46	58.71	68.42	57.69	67.99	59.83	69.13	55.51	66.83
FC-Siam-conc [19]	61.58	70.23	59.93	69.06	59.46	68.93	58.12	68.22	57.59	67.96	56.19	67.26
DSIFN [29]	88.72	89.25	88.42	89.02	87.47	88.21	85.29	86.41	84.12	85.45	83.16	84.71
ChangeFormer [24]	89.11	89.59	88.40	89.00	87.75	88.44	86.15	87.10	85.28	86.39	83.83	85.23
BIT [23]	90.29	90.63	89.89	90.29	88.12	88.76	86.21	87.15	85.50	86.58	84.27	85.58
DMFANet [47]	92.53	92.66	92.18	92.33	91.74	91.93	90.49	90.81	88.64	89.20	88.14	88.77
SUNet [37]	91.78	91.97	91.25	91.48	91.07	91.33	88.03	88.68	86.19	87.14	84.34	85.64
SRCDNet [9]	92.20	92.35	91.78	91.96	91.21	91.46	88.90	89.42	87.95	88.62	86.88	87.72
SILI [10]	92.49	92.62	92.30	92.43	91.65	91.86	90.12	90.49	88.59	89.16	88.09	88.73
DCILNet [48]	92.86	92.96	92.73	92.83	91.90	92.08	90.81	91.10	89.11	89.61	88.20	88.83
RDENet(proposed)	**93.72**	**93.76**	**93.56**	**93.60**	**92.94**	**93.04**	**91.18**	**91.43**	**90.62**	**90.93**	**89.17**	**89.66**

TABLE X

QUANTITATIVE COMPARISON WITH OTHER METHODS UNDER DIFFERENT RESOLUTION-DIFFERENCE CONDITIONS ON LEVIR-CD DATASET. (ADOPTING THE REVERSED CROSS-RESOLUTION CONFIGURATION, I.E., DOWNSAMPLING T1 WHILE MAINTAINING T2)

Method	4×		6×		7×		8×		9×		10×	
	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)	F1(%)	mIoU(%)
FC-EF [19]	74.25	78.25	73.41	77.64	72.27	76.90	71.73	76.52	67.88	74.07	66.41	73.13
FC-Siam-diff [19]	75.93	79.41	72.72	77.21	71.77	76.57	70.06	75.40	69.04	74.85	67.28	73.64
FC-Siam-conc [19]	72.52	77.12	72.10	76.81	71.23	76.22	70.34	75.64	68.29	74.38	66.35	73.15
DSIFN [29]	84.38	85.69	84.15	85.50	83.63	85.07	83.14	84.69	81.20	83.20	81.02	83.06
ChangeFormer [24]	85.29	86.43	84.93	86.14	84.05	85.41	82.37	84.09	81.14	83.14	78.97	81.53
BIT [23]	86.41	87.33	85.40	86.51	84.71	85.94	83.11	84.66	81.35	83.34	79.83	82.16
DMFANet [47]	90.26	90.61	89.73	90.14	88.13	88.76	86.38	87.30	85.69	86.73	84.89	86.08
SUNet [37]	85.29	86.42	84.12	85.47	83.15	84.70	82.29	84.04	81.45	83.40	80.24	82.49
SRCDNet [9]	84.98	86.18	84.33	85.63	83.90	85.30	82.58	84.26	82.18	83.95	81.27	83.25
SILI [10]	88.20	88.83	87.76	88.45	86.72	87.58	86.20	87.16	85.98	86.97	85.11	86.28
DCILNet [48]	90.45	90.78	90.02	90.40	88.25	88.87	87.29	88.05	85.91	86.91	85.51	86.58
RDENet(proposed)	**91.18**	**91.43**	**90.12**	**90.49**	**90.41**	**90.75**	**89.70**	**90.13**	**88.64**	**89.21**	**88.23**	**88.86**

conditions, with the ratio of high to LR set to 1:0.1311. The experimental results are recorded in Table XII.

As shown in the results of Table XII, under nonstandard resolution-difference conditions, RDENet still demonstrates the best performance. When the HR/LR ratio is 1:0.1311, RDENet achieves F1-score improvements of 1.10%, 0.99%, and 3.62% over the current cross-resolution method SILI on the three datasets, respectively. This indicates that the proposed RDENet is not only capable of handling integer-multiple resolution-difference conditions but also applicable to decimal-multiple resolution differences, making it suitable for a variety of cross-resolution scenarios.

4) Visualization Analysis: To intuitively demonstrate the advantages of RDENet in addressing cross-resolution CD tasks, we present two representative samples from the CDD(7×) dataset in Fig. 13. Through visualizing the output features of the model decoder, we analyze the mechanism behind the performance improvement of the proposed method in cross-resolution CD scenarios. The results demonstrate that RDENet effectively guides the model's attention to changed landcovers by integrating features extracted from bitemporal encoders and an image sequence encoder, thereby successfully suppressing task-irrelevant interference in complex cross-resolution scenarios.

As shown in the left sample, the LR image suffers from significant loss of spatial details in vehicles and interference from surrounding shadows, leading to missed detection in the baseline method. In contrast, RDENet can alleviate the problem of large spatial detail differences between bitemporal images by constructing pseudo-image sequences, and enhance the key features of changed objects through the resolution-difference modulation method, enabling accurate extraction of vehicle features and achieving precise detection results. The right sample demonstrates that newly constructed buildings, affected by seasonal variations and sharing similar radiative characteristics with surrounding snow, cause incomplete detection areas in the baseline method. However, RDENet effectively overcomes irrelevant interference by extracting distinctive building features to obtain complete detection results. In conclusion, the high concentration of RDENet's

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Fig. 12. Visual comparison on the cross-resolution LEVIR-CD dataset with varying resolution-difference ratios. (a) T1-temporal image (HR). (b) T2-temporal image (LR). (c) Ground truth. (d) DSIFN. (e) ChangeFormer. (f) BIT. (g) DMFANet. (h) SUNet. (i) SRCDNet. (j) SILI. (k) DCILNet. (l) Proposed RDENet.

TABLE XI
QUANTITATIVE COMPARISON WITH OTHER METHODS UNDER EQUAL-RESOLUTION CONDITION

Method	CDD				WHU-CD				LEVIR-CD			
	F1(%)	mIoU(%)	Kappa(%)	OA(%)	F1(%)	mIoU(%)	Kappa(%)	OA(%)	F1(%)	mIoU(%)	Kappa(%)	OA(%)
FC-EF [19]	85.69	85.71	83.89	96.80	61.18	70.09	59.20	96.22	75.68	79.25	74.48	97.70
FC-Siam-diff [19]	86.90	86.78	85.23	97.04	66.27	73.03	64.49	96.62	77.93	80.81	76.82	97.87
FC-Siam-conc [19]	87.73	87.51	86.15	97.19	67.54	73.82	65.83	96.76	76.82	80.05	75.68	97.81
DSIFN [29]	92.17	91.72	91.14	98.17	90.12	90.48	89.59	98.99	86.27	87.22	85.56	98.64
ChangeFormer [24]	92.97	92.53	92.06	98.38	90.98	91.24	90.50	99.08	86.83	87.68	86.14	98.69
BIT [23]	93.48	93.03	92.63	98.49	91.76	91.96	91.33	99.18	87.99	88.65	87.36	98.80
DMFANet [47]	94.88	94.47	94.22	98.82	92.93	93.03	92.56	99.29	90.51	90.83	90.00	99.04
SUNet [37]	93.02	92.58	92.12	98.41	92.29	92.44	91.88	99.23	88.10	88.75	87.48	98.82
SRCDNet [9]	94.87	94.45	94.20	98.81	92.78	92.89	92.40	99.28	87.69	88.40	87.05	98.77
SILI [10]	95.17	94.77	94.54	98.89	93.10	93.19	92.74	99.32	89.21	89.70	88.64	98.92
DCILNet [48]	95.41	95.02	94.81	98.95	93.18	93.26	92.82	99.32	90.84	91.12	90.35	99.07
RDENet(proposed)	**96.93**	**96.62**	**96.53**	**99.29**	**93.94**	**93.97**	**93.62**	**99.39**	**91.79**	**91.98**	**91.35**	**99.17**

attention on changed areas validates the overall effectiveness of our proposed components in addressing cross-resolution CD challenges.

In addition, we have observed that in the attention maps of the right sample, there are slight discrepancies between the regions of attention and the actual changed areas, and the fork in the road near the buildings is also highlighted. We believe the primary cause of this phenomenon is that the attention mechanism in RDENet not only focuses on direct changed targets (e.g., buildings) but also integrates contextual information (e.g., surrounding roads). While, this integration results in broader attention regions in the features. Furthermore, the attention maps represent intermediate visualizations from the feature extraction phase of RDENet, rather than the final output. The network's decoder and classifier modules further refine these features to precisely localize the changed regions. In future research, we aim to explore methods to guide the network to focus more precisely on changed regions in cross-resolution scenarios.

V. ABLATION STUDY

To systematically validate the impact of each module in the proposed RDENet on detection performance under cross-resolution scenarios, we conducted a series of ablation studies and hyperparameter sensitivity analyses. In this section, we first present the findings from componentwise ablation experiments, followed by a sensitivity analysis of key hyperparameters.

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TABLE XII

QUANTITATIVE COMPARISON WITH OTHER METHODS UNDER NONSTANDARD RESOLUTION-DIFFERENCE CONDITION. (THE HR/LR RATIO IS 1:0.1311, CORRESPONDING TO A 7.6278× DIFFERENCE)

Method	CDD				WHU-CD				LEVIR-CD			
	F1(%)	mIoU(%)	Kappa(%)	OA(%)	F1(%)	mIoU(%)	Kappa(%)	OA(%)	F1(%)	mIoU(%)	Kappa(%)	OA(%)
FC-EF [19]	79.09	80.18	76.50	95.39	52.73	65.68	50.46	95.66	72.06	76.72	70.60	97.23
FC-Siam-diff [19]	79.43	80.44	76.87	95.44	58.18	68.49	56.12	96.07	70.43	75.63	68.86	97.02
FC-Siam-conc [19]	81.21	81.89	78.86	95.82	58.87	68.93	56.91	96.26	70.66	75.84	69.16	97.15
DSIFN [29]	84.79	84.88	82.83	96.53	86.07	87.11	85.41	98.73	83.21	84.75	82.33	98.33
ChangeFormer [24]	88.26	87.99	86.73	97.30	86.73	87.65	86.09	98.78	82.97	84.55	82.07	98.28
BIT [23]	89.12	88.80	87.71	97.51	87.24	88.08	86.63	98.84	83.67	85.10	82.80	98.35
DMFANet [47]	93.27	92.83	92.40	98.46	91.04	91.35	90.61	99.18	87.23	88.00	86.55	98.70
SUNet [37]	91.99	91.57	90.97	98.18	89.03	89.59	88.51	99.00	82.50	84.21	81.60	98.28
SRCDNet [9]	93.27	92.83	92.40	98.46	90.22	90.63	89.75	99.11	82.74	84.38	81.84	98.28
SILI [10]	93.92	93.49	93.13	98.61	90.78	91.12	90.34	99.17	86.43	87.35	85.72	98.65
DCILNet [48]	93.50	93.06	92.66	98.51	91.13	91.43	90.70	99.19	87.94	88.60	87.30	98.78
RDENet(proposed)	**95.02**	**94.62**	**94.38**	**98.86**	**91.77**	**92.01**	**91.38**	**99.25**	**90.05**	**90.43**	**89.53**	**99.00**

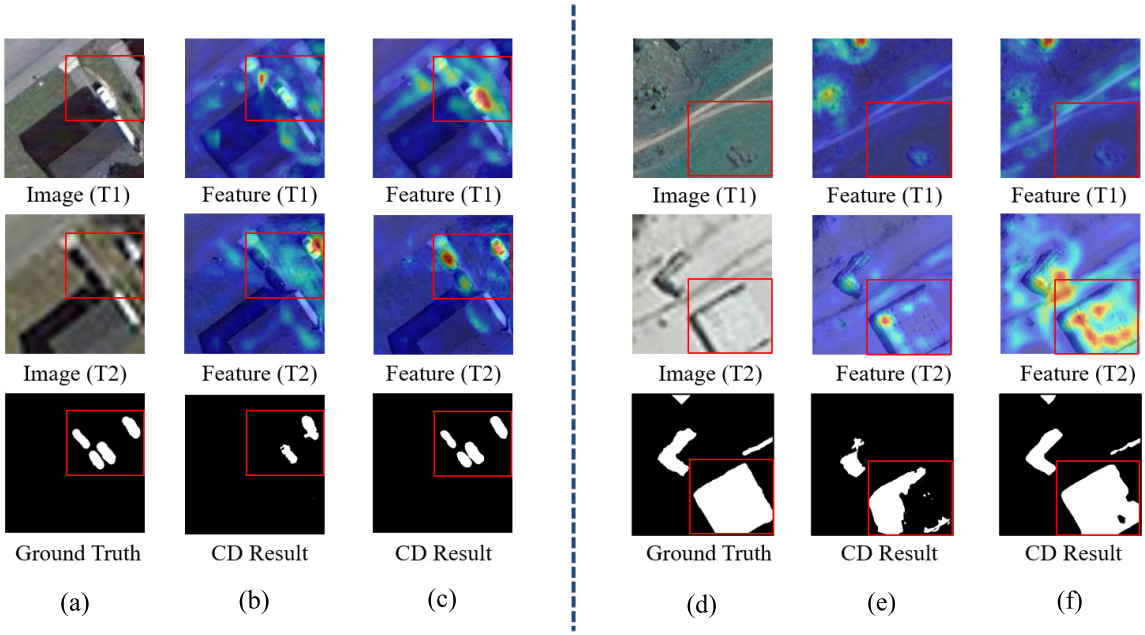


Fig. 13. Feature visualization of the proposed RDENet. (a) and (d) Bitemporal images and the ground truth (Original images). Features and CD results of (b) and (e) baseline and (c) and (f) RDENet.

A. *Effectiveness of Each Module*

To demonstrate the effectiveness of the proposed PRSR and BRDM (comprising RASM and RAMM), Table XIII presents ablation study results. The baseline configuration is defined as a CD model containing the bitemporal encoder and decoder without auxiliary designs. PRSR denotes pseudo-continuous resolution sequence representation method, while RASM and RAMM, respectively, represent resolution-aware self-modulation and resolution-aware mutual-modulation within the BRDM method.

The experimental results demonstrate progressive performance improvements when individually integrating the PRSR, RASM, and RAMM modules with the baseline configuration. Notably, the PRSR module achieves 1.10%, 0.98%, and 1.65% F1-score enhancement on the three datasets, respectively, validating the critical role of pseudo-image sequence encoding in cross-resolution CD. Furthermore, pairwise combinations of these modules yield superior performance compared with individual implementations, indicating a positive interaction of our proposed methods. Most significantly, the integration of all three modules culminates in peak performance, improving baseline's F1-scores of 2.21%, 1.77%, and 3.51% on the three datasets, respectively. This evidence substantiates that our RDENet effectively harnesses the complementarity of the pseudo-image sequence encoder and bitemporal encoder,

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TABLE XIII
ABLATION STUDY CONDUCTED ON THE PROPOSED RDENET

PRSR	RASM	RAMM	CDD(7×)				WHU-CD(6×)				LEVIR-CD(4×)			
			F1(%)	mIoU(%)	Kappa(%)	OA(%)	F1(%)	mIoU(%)	Kappa(%)	OA(%)	F1(%)	mIoU(%)	Kappa(%)	OA(%)
			93.17	92.74	92.29	98.45	91.71	91.90	91.26	99.14	87.76	88.45	87.12	98.77
✔			94.27	93.85	93.53	98.69	92.69	92.79	92.30	99.25	89.41	89.87	88.85	98.94
	✔		93.81	93.38	93.01	98.58	92.03	92.19	91.60	99.19	88.86	89.39	88.27	98.88
		✔	93.77	93.34	92.96	98.57	92.27	92.41	91.85	99.21	88.59	89.16	87.99	98.86
✔	✔		95.06	94.66	94.42	98.87	93.11	93.19	92.75	99.31	90.61	90.92	90.11	99.06
✔		✔	95.11	94.71	94.48	98.88	93.05	93.14	92.68	99.30	90.29	90.64	89.78	99.02
	✔	✔	94.90	94.49	94.24	98.83	92.97	93.06	92.60	99.30	90.44	90.77	89.94	99.04
✔	✔	✔	**95.38**	**94.99**	**94.78**	**98.94**	**93.48**	**93.53**	**93.13**	**99.33**	**91.27**	**91.51**	**90.81**	**99.12**

TABLE XIV
ABLATION STUDY CONDUCTED ON THE RAMM MODULE. (QKV_{BASE} AND QKV_{RD} REPRESENT THE CONVENTIONAL AND THE RESOLUTION-DIFFERENCE EMBEDDED Q–K–V COMPUTATION, RESPECTIVELY)

QKV_{base}	QKV_{rd}	CDD(7×)				WHU-CD(6×)				LEVIR-CD(4×)			
		F1(%)	mIoU(%)	Kappa(%)	OA(%)	F1(%)	mIoU(%)	Kappa(%)	OA(%)	F1(%)	mIoU(%)	Kappa(%)	OA(%)
		95.06	94.66	94.42	98.87	93.11	93.19	92.75	99.31	90.61	90.92	90.11	99.06
✔		95.21	94.81	94.59	98.90	93.29	93.35	92.93	99.32	91.03	91.30	90.56	99.10
	✔	95.28	94.88	94.67	98.92	93.35	93.41	92.99	**99.33**	91.11	91.37	90.64	99.11
✔	✔	**95.38**	**94.99**	**94.78**	**98.94**	**93.48**	**93.53**	**93.13**	**99.33**	**91.27**	**91.51**	**90.81**	**99.12**

thereby extracting discriminative features of changed landcovers and strengthening the model's change representation capability.

As demonstrated in the experimental results, when PRSR is activated individually, the F1-score improves by 1.10% on the CDD dataset; when BRDM (i.e., the combination of RASM and RAMM) is activated alone, the F1-score increases by 1.73% on the CDD dataset. When both PRSR and BRDM are fully activated, the F1-score achieves a 2.21% improvement. From the quantitative results, the gain achieved by combining PRSR and BRDM is slightly lower than the sum of their individual gains. The underlying reason for this phenomenon is that both PRSR and BRDM are designed to extract landcover features in the model: PRSR focuses on spatiotemporal features from image sequences, while BRDM emphasizes enhancing bitemporal key features. The redundancy between the landcover features extracted by these two methods prevents the gain from achieving linear improvement. Nevertheless, the two modules are complementary in enhancing the landcover features, and thus their combination contributes positively to the accurate detection of landcover changes, thereby improving the model's CD capability in cross-resolution scenarios.

Furthermore, since the RAMM module includes two branches: the conventional Q–K–V computation and the resolution-difference embedded Q–K–V computation, ablation experiments on the RAMM module are conducted, as shown in Table XIV. The experimental results show that both the conventional and the resolution-difference embedded Q–K–V can achieve cross-temporal feature interaction, leading to an improvement in CD performance. For example, on the CDD dataset, the conventional and resolution-difference embedded Q–K–V improved the F1-score by 0.16% and 0.22%, respectively. Their combination further enhances the detection performance, achieving F1-score improvements of 0.32%, 0.37%, and 0.66% on the three datasets, respectively, which demonstrates the effectiveness of the two computational branches.

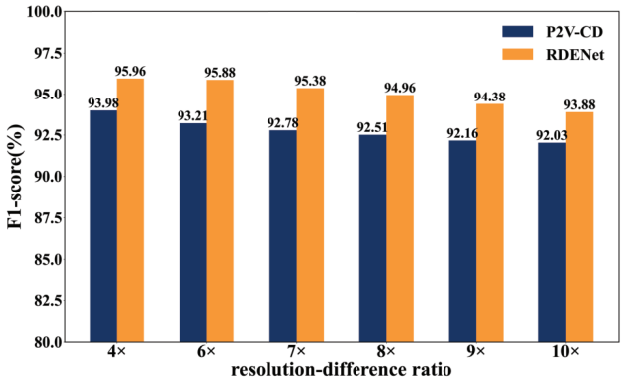


Fig. 14. Comparison between RDENet and P2V-CD on the cross-resolution CDD dataset.

B. Comparison With Existing Pseudo-Image Sequence Construction Method

The proposed RDENet adopts a PRSR method. In existing research, image sequence modeling approaches have been applied to the RS field, such as P2V-CD [41]. Therefore, to fairly evaluate the superiority and advancement of the proposed method, we conducted extensive comparisons between RDENet and P2V-CD on three datasets. The experimental results are presented in Figs. 14–16.

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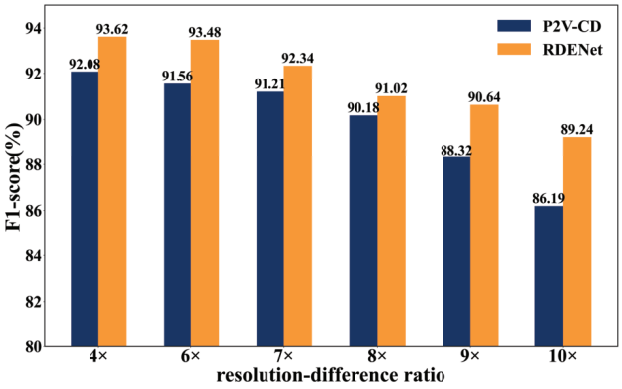


Fig. 15. Comparison between RDENet and P2V-CD on the cross-resolution WHU-CD dataset.

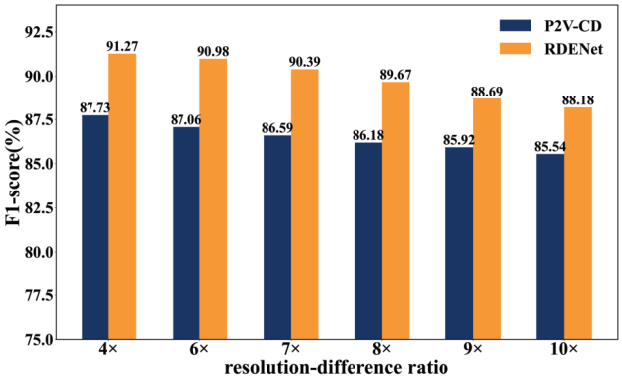


Fig. 16. Comparison between RDENet and P2V-CD on the cross-resolution LEVIR-CD dataset.

As shown in the figures, the proposed RDENet outperforms P2V-CD in terms of performance across all three datasets. For instance, under a 4× resolution-difference setting, the F1-scores of RDENet are 1.98%, 1.54%, and 3.54% higher, respectively, compared with P2V-CD. The superiority of RDENet primarily stems from its design of pseudo-image sequences tailored for cross-resolution scenarios and the introduction of a RGFF method, which enhances its adaptability to such scenarios. In contrast, P2V-CD is a conventional image sequence modeling method that lacks specialized designs for cross-resolution real scenarios, leading to poorer performance compared with RDENet.

Furthermore, as the resolution-difference ratio increases, the performance of both RDENet and P2V-CD decreases, primarily due to the loss of more spatial details in RS images, which makes the identification of changed landcovers more challenging. However, under all resolution-difference settings, RDENet consistently outperforms P2V-CD, demonstrating its superior performance in cross-resolution scenarios.

C. Cross-Resolution Generalization Experiments

In this section, a series of cross-resolution generalization experiments are conducted to validate the robustness of RDENet. Specifically, we trained the model on 4× resolution-difference image pairs and evaluated its performance across resolution differences of 6×, 7×, 8×, 9×, and 10× spanning three datasets. The results are presented as visual line charts in Figs. 17–19.

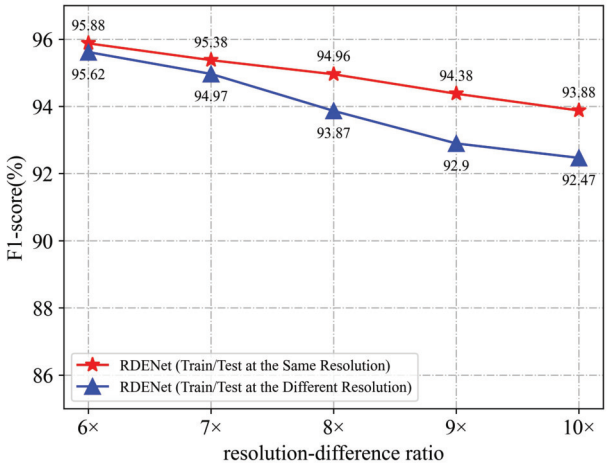


Fig. 17. Cross-resolution generalization results on the CDD dataset.

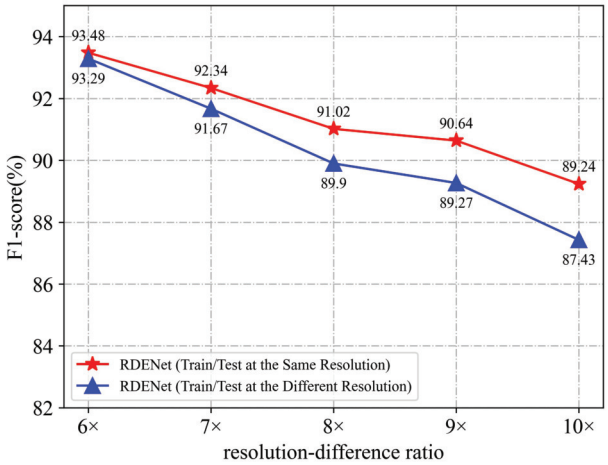


Fig. 18. Cross-resolution generalization results on the WHU-CD dataset.

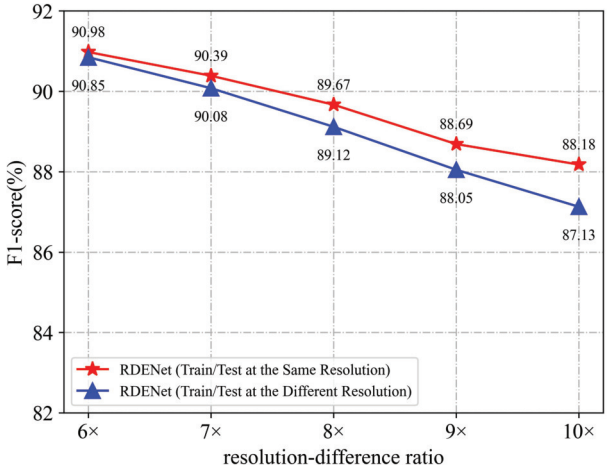


Fig. 19. Cross-resolution generalization results on the LEVIR-CD dataset.

Overall, compared with detection performance when training and testing use the same resolution difference, the F1-score results under varying resolution differences are slightly lower. The primary reason for this performance degradation is that test images lose more detailed information compared

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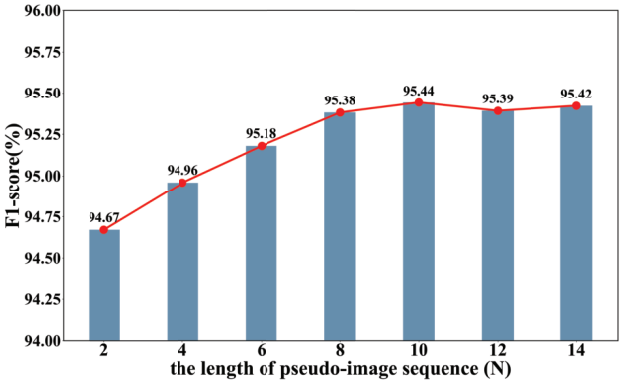


Fig. 20. Trend of the $F1$-score with respect to different values of N.

TABLE XV

EFFECT OF THE LENGTH OF PSEUDO-IMAGE SEQUENCE ON THE CROSS-RESOLUTION CDD(7×) DATASET

N	FLOPs(G)	F1(%)	mIoU(%)	Kappa(%)	OA(%)
2	23.32	94.67	94.26	93.98	98.78
4	25.32	94.96	94.55	94.31	98.84
6	27.59	95.18	94.78	94.55	98.89
8	29.58	95.38	94.99	94.78	98.94
10	31.85	95.44	95.05	94.85	98.95
12	33.84	95.39	95.00	94.79	98.94
14	36.11	95.42	95.03	94.83	98.95

with the training set. However, by adopting the resolution-difference feature modulation method in RDENet, the model exhibits adaptive generalization capabilities across diverse resolution-difference scenarios. Consequently, even under a 10× resolution difference, the precision loss in F1-score remains within 2%, demonstrating the superiority of the proposed RDENet.

D. Hyperparameters Analysis

In this section, we perform hyperparameter sensitivity analysis to investigate the effect of the length N of the pseudo-image sequence constructed in PRSR on the performance of RDENet. To this end, we selected different N values and conducted experiments on the CDD(7×) dataset to record the performance of the model under different N values, as shown in Table XV. For a more intuitive visualization, we also plot the trend of the F1-score with respect to different values of N in Fig. 20. It is obvious that with the increase of N value, the F1-score gradually increases until it becomes stable. This is consistent with the mechanism of PRSR, which can help the model to achieve spatiotemporal modeling under the condition of cross resolution by constructing pseudo-sequence images. However, when $N > 8$, the performance of the model tends to be stable, and the F1-score is no longer significantly improved, or even slightly decreased. This shows that when the constructed pseudo-image sequence is too long, the redundant information will have a negative effect on the model learning and lead to the increase of the model parameter size and the increase of the calculation cost. An appropriate value of N will achieve a balance between model performance improvement and computational overhead, and experimental results show that $N = 8$ is an appropriate choice.

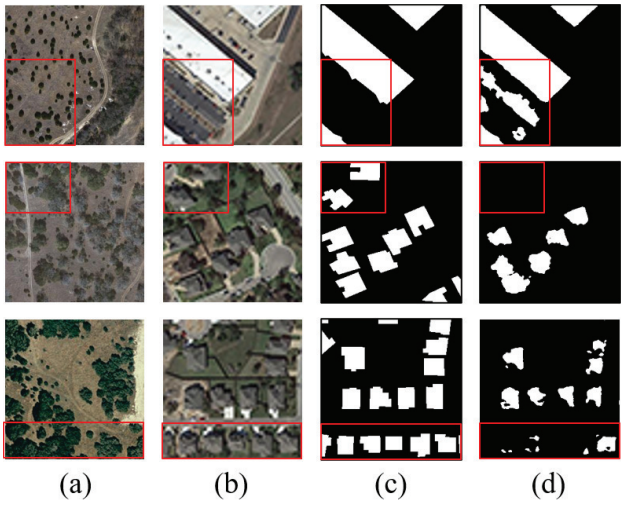


Fig. 21. Visualization examples of RDENet trained on the WHU-CD(6×) dataset and tested on the LEVIR-CD(4×) dataset. (a) T1-temporal image (HR). (b) T2-temporal image (LR). (c) Ground truth. (d) CD Results.

TABLE XVI

GENERALIZATION EXPERIMENT RESULTS ACROSS DATASETS

Training dataset → Test dataset	F1(%)	mIoU(%)	Kappa(%)	OA(%)
LEVIR-CD → LEVIR-CD	91.27	91.51	90.81	99.12
WHU-CD → WHU-CD	93.48	93.53	93.13	99.33
WHU-CD → LEVIR-CD	49.56	64.56	47.89	96.25
LEVIR-CD → WHU-CD	65.68	72.41	63.64	96.08

VI. DISCUSSION

The proposed RDENet demonstrates excellent detection performance in cross-resolution CD tasks. Furthermore, we explored the limitations of RDENet by evaluating its performance in cross-domain generalization scenarios. We conducted cross-domain experiments on the WHU-CD(6×) and LEVIR-CD(4×) datasets. First, the proposed RDENet model was trained on the WHU-CD dataset and tested on the LEVIR-CD dataset. Then, we trained on the LEVIR-CD dataset and tested on the WHU-CD dataset. Quantitative experimental results are recorded in Table XVI, and some visualization examples are shown in Fig. 21.

As shown in Table XVI, the proposed RDENet fails to achieve strong cross-domain generalization when the training and test sets belong to different datasets. When trained on WHU-CD and tested on LEVIR-CD, the F1-score decreases by 41.71% compared with training and testing on LEVIR-CD itself. Testing on WHU-CD results in a 27.80% drop in F1-score. These quantitative results indicate that the current RDENet struggles with CD tasks across different datasets. The visualizations in Fig. 21 further support this conclusion, showing numerous false alarms and missed detections during cross-domain testing. These findings and analyses reveal the limitations of RDENet and highlight an important direction for our future work.

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VII. Conclusion

This article proposes RDENet, which integrates a PRSR method and a BRDM method to enhance the model's perception capability for changed landcovers in complex scenarios under cross-resolution conditions. The proposed PRSR method constructs pseudo-image sequences with continuous resolution transitions and designs a RGFF module to alleviate the difficulty in extracting spatial detail features of landcovers caused by resolution difference. The BRDM method introduces a resolution-difference factor into the learning process through RASM and RAMM modules, guiding the model to adaptively enhance key features under different resolutions. This enhances the identification of changed and unchanged regions and improves cross-resolution CD performance. The proposed RDENet has demonstrated effectiveness across multiple datasets. Both quantitative and visual experimental results indicate that RDENet is suitable for CD tasks involving various resolution-difference conditions.

In future work, we plan to explore combining domain generalization technologies with the proposed cross-resolution methods to develop robust CD approaches capable of addressing varying resolution differences and varying scenarios [49]. On the other hand, emerging advanced CD models have demonstrated strong performance [50], [51], [52], [53], while certain vision foundation models have also shown powerful representational capabilities [54], [55], [56]. Exploring their potential role in the context of integrating domain adaptation with cross-resolution CD is a key avenue for our future research.

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